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Inventor(s)	Wheeler; Colin et al.

Structural refuse vehicle body with undermounted batteries

Abstract

An electrified vehicle includes a front subframe, a rear subframe, and a body that connects the front subframe to the rear subframe so that forces acting on the front subframe and the rear subframe are transmitted through the body. The vehicle also includes a housing coupled to an underside of the body between the front subframe and the rear subframe. An energy storage system is enclosed in the housing and includes a battery and a battery interface configured to electrically couple the battery to at least one component of the electrified vehicle.

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Office Action issued in connection with Canadian Appl. No. 3173738 dated Oct. 17, 2024. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This patent application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/242,304, filed on Sep. 9, 2021, and U.S. Provisional Patent Application No. 63/242,317, filed on Sep. 9, 2021, which are incorporated by reference herein in their entirety.

BACKGROUND

(1) The present disclosure relates generally to vehicles. More specifically, the present disclosure relates to a vehicle including a chassis. The chassis may be coupled to components, such as a body or an implement, that are specific to a desired application of the vehicle.

SUMMARY

(2) One exemplary embodiment of the invention relates to an electrified vehicle. The electrified vehicle includes a chassis, a body, and a cab. The chassis is configured to support a tractive element. The chassis includes a battery box. The body is supported by the chassis. The cab is supported by the chassis. The battery box includes a shell defining an internal cavity. The battery box is configured to receive a module. The module includes a battery and a module terminal. The battery box is configured to transfer energy from the module to at least one component of the electrified vehicle. The internal cavity of the battery box comprises a system terminal configured to contact the module terminal of the module to facilitate the transfer of the energy from the module to the component of the electrified vehicle.

(3) Another exemplary embodiment of the invention relates to an electrified vehicle. The electrified vehicle includes a chassis, a body, a cab, and a battery box. The chassis is configured to support a tractive element. The body is supported by the chassis. The cab is supported by the chassis. The battery box is coupled to the chassis and is configured to receive a module. The battery box comprises a first portion and a second portion. The first portion comprises a first depth and a first width and the second portion comprising a second depth and a second width. The first depth is different from the second depth and the first width different from the second width.

(4) Another exemplary embodiment of the invention relates to a method of providing energy to a component of a vehicle. The method includes receiving, by a chassis of the vehicle, a module comprising at least one electrical energy storage device configured to store the energy and a module terminal. The chassis is configured to support a tractive element. The chassis comprising a battery box. The battery box comprising a system terminal. The method includes transferring, by the chassis of the vehicle, energy from the electrical energy storage device of the module to the component of the vehicle. The energy transfers from the module via the module terminal and to the component via the system terminal of the battery box.

(5) Another exemplary embodiment relates to an electrified vehicle. The electrified vehicle includes a front subframe, a rear subframe, and a body. The body connects the front subframe to the rear subframe so that forces acting on the front subframe and the rear subframe are transmitted through the body. A housing is coupled to the underside of the body between the front subframe and the rear subframe. An energy storage system is enclosed within the housing and includes a battery and a battery interface configured to electrically couple the battery to at least one component of the electrified vehicle.

(6) Another exemplary embodiment relates to an electrified vehicle. The electrified vehicle includes a chassis. The chassis includes a front subframe, a rear subframe, and a body that directly couples the front subframe to the rear subframe. A tractive element is supported by the chassis. A drive motor is supported by the chassis and coupled to the tractive element to rotate the tractive

element. A housing is coupled to the body between the front subframe and the rear subframe. The housing includes a battery, and the battery is electrically coupled to the drive motor.

(7) The invention is capable of other embodiments and of being carried out in various ways. Alternative exemplary embodiments relate to other features and combinations of features as may be recited herein.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:
- (2) FIG. 1 is a left side view of a vehicle, according to an exemplary embodiment.
- (3) FIG. 2 is a perspective view of a chassis of the vehicle of FIG. 1.
- (4) FIG. 3 is a perspective view of the vehicle of FIG. 1 configured as a refuse vehicle, according to an exemplary embodiment.
- (5) FIG. 4 is a perspective view of the vehicle of FIG. 1 configured as a mixer vehicle, according to an exemplary embodiment.
- (6) FIG. 5 is a perspective view of the vehicle of FIG. 1 configured as a fire fighting vehicle, according to an exemplary embodiment.
- (7) FIG. 6 is a left side view of the vehicle of FIG. 1 configured as an airport fire fighting vehicle, according to an exemplary embodiment.
- (8) FIG. 7 is a perspective view of the vehicle of FIG. 1 configured as a boom lift, according to an exemplary embodiment.
- (9) FIG. 8 is a perspective view of the vehicle of FIG. 1 configured as a scissor lift, according to an exemplary embodiment.
- (10) FIG. 9A is a side view of a chassis of the vehicle of FIG. 1, according to an exemplary embodiment.
- (11) FIG. 9B is a top view of a chassis of the vehicle of FIG. 1, according to an exemplary embodiment.
- (12) FIG. 10 is a top view of a chassis of the vehicle of FIG. 1, according to another exemplary embodiment.
- (13) FIG. 11A is a top view of an internal cavity of a battery box of the vehicle of FIG. 1, according to an exemplary embodiment.
- (14) FIG. 11B is a side view of the internal cavity of the battery box of FIG. 11A, according to an exemplary embodiment.
- (15) FIG. 12A is a top view of an internal cavity of a battery box of the vehicle of FIG. 1, according to another exemplary embodiment.
- (16) FIG. 12B is a side view of an internal cavity of a battery box of the vehicle of FIG. 1, according to another exemplary embodiment.
- (17) FIG. 13A is a side view of a battery box of the vehicle of FIG. 1, according to an exemplary embodiment.
- (18) FIG. 13B is a side view of a battery box of the vehicle of FIG. 1, according to another exemplary embodiment.
- (19) FIG. 13C is a side view of a battery box of the vehicle of FIG. 1, according to another exemplary embodiment.
- (20) FIG. 14A is a schematic diagram of an energy transfer from a module to a component of the vehicle of FIG. 1, according to an exemplary embodiment.
- (21) FIG. 14B is a schematic diagram of an energy transfer from a module to a component of the

vehicle of FIG. 1, according to another exemplary embodiment.

(22) FIG. 15 is a flow diagram of a method of providing energy to at least one component of the vehicle of FIG. 1, according to an exemplary embodiment.

(23) FIG. 16 is a schematic diagram of a module of the vehicle of FIG. 1, according to an exemplary embodiment.

(24) FIG. 17 is a schematic diagram of a power distribution unit of the vehicle of FIG. 1, according to an exemplary embodiment.

(25) FIG. 18 is a schematic diagram of a battery box of the vehicle of FIG. 1, according to an exemplary embodiment.

(26) FIG. 19 is a left side view of the vehicle of FIG. 1 configured with a body connecting two subframes, according to an exemplary embodiment.

(27) FIG. 20 is a bottom view of the vehicle of FIG. 19, according to an exemplary embodiment.

(28) FIG. 21 is a side view of a battery housing of the vehicle of FIG. 19, according to an exemplary embodiment.

(29) FIG. 22 is a bottom view of the vehicle of FIG. 19 configured with a three-point mounting system, according to an exemplary embodiment.

(30) FIG. 23 is a bottom view of the vehicle of FIG. 19 configured with a four-point mounting system, according to an exemplary embodiment.

(31) FIG. 24 is a top view of a battery housing of the vehicle of FIG. 19, according to an exemplary embodiment.

(32) FIG. 25 is a cross-sectional view of the battery housing of FIG. 24 taken along line A-A.

DETAILED DESCRIPTION

(33) Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

(34) According to an exemplary embodiment, vocational vehicles (e.g., refuse trucks, mixing vehicles) include a battery box that defines a structural component of a chassis of the vehicle. The battery box provides support and structure for the vehicle while also providing storage, protection, functionality, and accessibility for batteries and other systems. In one embodiment, the battery box is disposed on the bottom of a vocational vehicle between a front and a rear axle. The battery box may have integrated suspension hard points. In other embodiments, a front or a rear subframe is attached to the battery box. The battery box may extend the full width of the vehicle to increase the volume for batteries and other systems therein. In other embodiments, the battery box extends over the rear axle(s). Such an embodiment leads to improved weight distribution and has increased battery volume. In one embodiment, the battery box includes battery mounting provisions, a system routing pathway, a bus bar, a thermal management system, a power distribution unit, and a charge port, among other components and systems.

(35) Incorporating the battery box as a structural element of a vocational vehicle simplifies installation and maintenance of batteries and other vehicle systems and improves the functionality of the vehicles. According to an exemplary embodiment, the battery box as described herein provides an alternative power source to the traditional fuel source, saves space inside the vehicle by moving the batteries and other systems to the exterior, reduces weight by eliminating redundant structures, and simplifies assembly, repair, and replacement by allowing the systems to be assembled off the truck then installed as a complete module, among others.

(36) According to another exemplary embodiment an electrified vocational vehicle (e.g., refuse truck, mixer vehicle, fire fighting vehicle, etc.) includes a body that is a main structural component thereof, thereby forming a structural body. The electrified vocational vehicle may not have a conventional frame (e.g., a pair of frame rails). In one embodiment, the body has an interchangeable energy storage system positioned where the frame would traditionally be

positioned. The structural body eliminates conventional components and reduces weight. Further, on-vehicle space previously occupied by the frame is now available to house other vehicle components such as the interchangeable energy storage system. In one embodiment, the structural body connects front and rear subframes of the vehicle as the main structural member of the chassis, the energy storage system for powering the vehicle and/or its various components, a cab either directly or through the front subframe, etc., to provide support as the main structural member for the entire vehicle. In one embodiment, the energy storage system includes a battery housing that receives and protects an interchangeable battery module. The battery module includes the batteries, associated cabling, thermal management systems, communication systems, monitoring systems, etc., and provides a complete, swappable, battery module that can be assembled independent of the vehicle and installed as a complete unit. The vehicle with the structural body acting as the includes an energy storage system mounted to the underside of the body where a conventional frame would otherwise be located, according to an exemplary embodiment.

(37) According to an exemplary embodiment, as shown in FIGS. 1 and 2, a vehicle (e.g., a vehicle assembly, a truck, a vehicle base, etc.), shown as vehicle **10**, includes a frame assembly or chassis assembly, shown as chassis **20**. The chassis assembly may support other components of the vehicle **10**. In some embodiments, the chassis **20** extends longitudinally along a length of the vehicle **10**. The chassis **10** may extend substantially parallel to a primary direction of travel of the vehicle **10**. According to an exemplary embodiment, the chassis **20** includes three sections or portions, shown as front section **22**, middle section **24**, and rear section **26**. The middle section **24** of the chassis **20** extends between the front section **22** and the rear section **26**. In some embodiments, the middle section **24** of the chassis **20** couples the front section **22** to the rear section **26**. In other embodiments, the front section **22** is coupled to the rear section **26** by another component (e.g., the body of the vehicle **10**).

(38) As shown in FIG. 2, the front section **22** includes a pair of frame portions, frame members, or frame rails, shown as front rail portion **30** and front rail portion **32**. The rear section **26** includes a pair of frame portions, frame members, or frame rails, shown as rear rail portion **34** and rear rail portion **36**. The front rail portion **30** is laterally offset from the front rail portion **32**. Similarly, the rear rail portion **34** is laterally offset from the rear rail portion **36**. This spacing provides frame stiffness and space for vehicle components (e.g., batteries, motors, axles, gears, etc.) between the frame rails. In some embodiments, the front rail portions **30** and **32** and the rear rail portions **34** and **36** extend longitudinally and substantially parallel to one another. The chassis **20** may include additional structural elements (e.g., cross members that extend between and couple the frame rails).

(39) In some embodiments, the front section **22** and the rear section **26** are configured as separate, discrete subframes (e.g., a front subframe and a rear subframe). In such embodiments, the front rail portion **30**, the front rail portion **32**, the rear rail portion **34**, and the rear rail portion **36** are separate, discrete frame rails that are spaced apart from one another. In some embodiments, the front section **22** and the rear section **26** are each directly coupled to the middle section **24** such that the middle section **24** couples the front section **22** to the rear section **26**. Accordingly, the middle section **24** may include a structural housing or frame. In other embodiments, the front section **22**, the middle section **24**, and the rear section **26** are coupled to one another by another component, such as a body of the vehicle **10**.

(40) In other embodiments, the front section **22**, the middle section **24**, and the rear section **26** are defined by a pair of frame rails that extend continuously along the entire length of the vehicle **10**. In such an embodiment, the front rail portion **30** and the rear rail portion **34** would be front and rear portions of a first frame rail, and the front rail portion **32** and the rear rail portion **36** would be front and rear portions of a second frame rail. In such embodiments, the middle section **24** would include a center portion of each frame rail.

(41) In some embodiments, the middle section **24** acts as a storage portion that includes one or more vehicle components. The middle section **24** may include an enclosure that contains one or

more vehicle components and/or a frame that supports one or more vehicle components. In some embodiments, the middle section **24** contains or includes one or more electrical energy storage devices (e.g., batteries, capacitors, etc.). In another embodiment, the middle section **24** includes fuel tanks. In yet another embodiment, the middle section **24** defines a void space or storage volume that can be filled by a user.

(42) According to an exemplary embodiment, a cabin, operator compartment, or body component, shown as cab **40**, is coupled to a front end portion of the chassis **20** (e.g., the front section **22** of the chassis **20**). Together, the chassis **20** and the cab **40** define a front end of the vehicle **10**. The cab **40** extends above the chassis **20**. The cab **40** includes an enclosure or main body that defines an interior volume, shown as cab interior **42** that is sized to contain one or more operators. The cab **40** also includes one or more doors **44** that facilitate selective access to the cab interior **42** from outside of the vehicle **10**. The cab interior **42** contains one or more components that facilitate operation of the vehicle **10** by the operator. In one embodiment, the cab interior **42** contains components that facilitate operator comfort (e.g., seats, seatbelts, etc.), user interface components that receive inputs from the operators (e.g., steering wheels, pedals, touch screens, switches, buttons, levers, etc.), and/or user interface components that provide information to the operators (e.g., lights, gauges, speakers, etc.). The user interface components within the cab **40** may facilitate operator control over the drive components of the vehicle **10** and/or over any implements of the vehicle **10**.

(43) According to an exemplary embodiment, the vehicle **10** further includes a series of axle assemblies, shown as front axle **50** and rear axles **52**. As shown, the vehicle **10** includes one front axle **50** coupled to the front section **22** of the chassis **20** and two rear axles **52** each coupled to the rear section **26** of the chassis **20**. In other embodiments, the vehicle **10** includes more or fewer axles. In one embodiment, the vehicle **10** includes a tag axle that may be raised or lowered to accommodate variations in weight being carried by the vehicle **10**. The front axle **50** and the rear axles **52** each include a plurality of tractive elements (e.g., wheels, treads, etc.), shown as wheel and tire assemblies **54**. The wheel and tire assemblies **54** are configured to engage a support surface (e.g., roads, the ground, etc.) to support and propel the vehicle **10**. The front axle **50** and the rear axles may include steering components (e.g., steering arms, steering actuators, etc.), suspension components (e.g., gas springs, dampeners, air springs, etc.), power transmission or drive components (e.g., differentials, drive shafts, etc.), braking components (e.g., brake actuators, brake pads, brake discs, brake drums, etc.), and/or other components that facilitate propulsion or support of the vehicle.

(44) In some embodiments, the vehicle **10** is configured as an electric vehicle that is propelled by an electric powertrain system. As shown in FIG. **1**, the vehicle **10** includes one or more electrical energy storage devices (e.g., batteries, capacitors, etc.), shown as batteries **60**. As shown, the batteries **60** are positioned within the middle section **24** of the chassis **20**. In other embodiments, the batteries **60** are otherwise positioned throughout the vehicle **10**. The vehicle **10** further includes one or more electromagnetic devices (e.g., motor/generators), shown as drive motors **62**. The drive motors **62** are electrically coupled to the batteries **60**. The drive motors **62** may be configured to receive electrical energy from the batteries **60** and provide rotational mechanical energy to the wheel and tire assemblies **54** to propel the vehicle **10**. The drive motors **62** may be configured to receive rotational mechanical energy from the wheel and tire assemblies **64** and provide electrical energy to the batteries **60**, providing a braking force to slow the vehicle **10**. As shown, the drive motors **62** are positioned within the rear axles **52** (e.g., as part of a combined axle and motor assembly). In other embodiments, the drive motors **62** are otherwise positioned within the vehicle **10**.

(45) In other embodiments, the vehicle **10** is configured as a hybrid vehicle that is propelled by a hybrid powertrain system (e.g., a diesel/electric hybrid, gasoline/electric hybrid, natural gas/electric hybrid, etc.). According to an exemplary embodiment, the hybrid powertrain system includes a

primary driver (e.g., an engine, a motor, etc.), an energy generation device (e.g., a generator, etc.), and/or an energy storage device (e.g., a battery, capacitors, ultra-capacitors, etc.) electrically coupled to the energy generation device. The primary driver may combust fuel (e.g., gasoline, diesel, etc.) to provide mechanical energy, which a transmission may receive and provide the axle front axle **50** and/or the rear axles **52** to propel the vehicle **10**. Additionally or alternatively, the primary driver may provide mechanical energy to the generator, which converts the mechanical energy into electrical energy. The electrical energy may be stored in the energy storage device (e.g., the batteries **60**) in order to later be provided to a motive driver.

(46) In yet other embodiments, the chassis **20** is further be configured to support non-hybrid powertrains. For example, the powertrain system may include a primary driver that is a compression-ignition internal combustion engine that utilizes diesel fuel.

(47) As shown in FIG. **1**, the vehicle **10** includes a rear assembly, module, implement, body, or cargo area, shown as application kit **80**. The application kit **80** may include one or more implements, vehicle bodies, and/or other components. Although the application kit **80** is shown positioned behind the cab **40**, in other embodiments the application kit **80** extends forward of the cab **40**. The vehicle **10** may be outfitted with a variety of different application kits **80** to configure the vehicle **10** for use in different applications. Accordingly, a common vehicle **10** can be configured for a variety of different uses simply by selecting an appropriate application kit **80**. By way of example, the vehicle **10** may be configured as a refuse vehicle, a concrete mixer, a fire fighting vehicle, an airport fire fighting vehicle, a lift device (e.g., a boom lift, a scissor lift, a telehandler, a vertical lift, etc.), a crane, a tow truck, a military vehicle, a delivery vehicle, a mail vehicle, a boom truck, a plow truck, a farming machine or vehicle, a construction machine or vehicle, a coach bus, a school bus, a semi-truck, a passenger or work vehicle (e.g., a sedan, a SUV, a truck, a van, etc.), and/or still another vehicle. FIGS. **3-8** illustrate various examples of how the vehicle **10** may be configured for specific applications. Although only a certain set of vehicle configurations is shown, it should be understood that the vehicle **10** may be configured for use in other applications that are not shown.

(48) According to an exemplary embodiment, the application kit **80** includes various actuators to facilitate certain functions of the vehicle **10**. In one embodiment, the application kit **80** includes hydraulic actuators (e.g., hydraulic cylinders, hydraulic motors, etc.), pneumatic actuators (e.g., pneumatic cylinders, pneumatic motors, etc.), and/or electrical actuators (e.g., electric motors, electric linear actuators, etc.). The application kit **80** may include components that facilitate operation of and/or control of these actuators. In another embodiment, the application kit **80** includes hydraulic or pneumatic components that form a hydraulic or pneumatic circuit (e.g., conduits, valves, pumps, compressors, gauges, reservoirs, accumulators, etc.). By way of another embodiment, the application kit **80** includes electrical components (e.g., batteries, capacitors, voltage regulators, motor controllers, etc.). The actuators may be powered by components of the vehicle **10**. In some embodiments, the actuators are powered by the batteries **60**, the drive motors **62**, or the primary driver (e.g., through a power take off).

(49) As shown in FIG. **3**, the vehicle **10** is configured as a refuse vehicle **100** (e.g., a refuse truck, a garbage truck, a waste collection truck, a sanitation truck, a recycling truck, etc.). Specifically, the refuse vehicle **100** is a front-loading refuse vehicle. In other embodiments, the refuse vehicle **100** is configured as a rear-loading refuse vehicle or a side-loading refuse vehicle.

(50) As shown in FIG. **3**, the application kit **80** of the refuse vehicle **100** includes a rear body or container, shown as refuse compartment **130**, and a pivotable rear portion, shown as tailgate **132**. The refuse compartment **130** may facilitate transporting refuse from various waste receptacles within a municipality to a storage and/or a processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). According to an exemplary embodiment, loose refuse is placed into the refuse compartment **130** to be compacted. The refuse compartment **130** may also provide temporary storage for refuse during transport to a waste disposal site and/or a recycling facility. In

some embodiments, the refuse compartment **130** includes a hopper volume and storage volume. In this regard, refuse may be initially loaded into the hopper volume and later compacted into the storage volume. According to an exemplary embodiment, the hopper volume is positioned between the storage volume and the cab **40** (e.g., refuse is loaded into a position of the refuse compartment **130** behind the cab **40** and stored in a position further toward the rear of the refuse compartment **130**). In other embodiments, the storage volume is positioned between the hopper volume and the cab **40** (e.g., in a rear-loading refuse truck, etc.). The tailgate **132** may be pivotally coupled to the refuse compartment **130**, and may be movable between a closed position and an open position by an actuator (e.g., a hydraulic cylinder, an electric linear actuator, etc.), shown as tailgate actuator **134** (e.g., to facilitate emptying the storage volume).

(51) As shown in FIG. 3, the refuse vehicle **100** also includes an implement, shown as lift assembly **108** (e.g., a front-loading lift assembly, etc.). According to an exemplary embodiment, the lift assembly **108** includes a pair of lift arms **140**, lift arm actuators **142**, and articulation actuators **144**. The lift arms **140** may be rotatably coupled to the chassis **20**. In another embodiment, the lift arms **140** are rotatably coupled to the refuse compartment **30** on each side of the refuse vehicle **100** (e.g., through a pivot, a lug, a shaft, etc.). Such an embodiment provides that the lift assembly **108** extends forward relative to the cab **40** (e.g., a front-loading refuse truck, etc.). In other embodiments, the lift assembly **108** extends rearward relative to the application kit **80** (e.g., a rear-loading refuse truck). In yet other embodiments, the lift assembly **108** extends from a side of the application kit **80** (e.g., a side-loading refuse truck). As shown in FIG. 3, the lift arm actuators **142** are positioned such that extension and retraction of the lift arm actuators **142** rotates the lift arms **140** about an axis extending through the pivot. In this regard, the lift arms **140** may be rotated by the lift arm actuators **142** to lift a refuse container over the cab **40**. In an exemplary embodiment, the articulation actuators **144** are positioned to articulate the distal end of the lift arms **140** (e.g., a portion of the lift arms **140** that may be coupled to the refuse container) in order to assist in tipping refuse out of the refuse container and into the refuse compartment **130**. The lift arm actuators **142** may then rotate the lift arms **140** to return the empty refuse container to the ground.

(52) According to another exemplary embodiment, as shown in FIG. 4, the vehicle **10** is configured as a mixer truck (e.g., a concrete mixer truck, a mixer vehicle, etc.), shown as mixer truck **200**. Specifically, the mixer truck **200** is a rear-discharge concrete mixer truck. In other embodiments, the mixer truck **200** is a front-discharge concrete mixer truck.

(53) As shown in FIG. 4, the application kit **80** includes a mixing drum assembly (e.g., a concrete mixing drum), shown as drum assembly **230**. The drum assembly **230** includes a mixing drum **232**, a drum drive system **234** (e.g., a rotational actuator or motor), an inlet, shown as hopper **236**, and an outlet, shown as chute **238**. The mixing drum **232** may be coupled to the chassis **20** and may be disposed behind the cab **40** (e.g., at the rear and/or middle of the chassis **20**). In an exemplary embodiment, the drum drive system **234** is coupled to the chassis **20** and configured to selectively rotate the mixing drum **232** about a central, longitudinal axis. According to an exemplary embodiment, the central, longitudinal axis of the mixing drum **232** is elevated from the chassis **20** (e.g., from a horizontal plan extending along the chassis **20**) at an angle in the range of five degrees to twenty degrees. In other embodiments, the central, longitudinal axis is elevated by less than five degrees (e.g., four degrees, etc.). In yet another embodiment, the mixer truck **200** includes an actuator positioned to facilitate adjusting the central, longitudinal axis to a desired or target angle (e.g., manually in response to an operator input/command, automatically according to a control system, etc.).

(54) The mixing drum **232** may be configured to receive a mixture, such as a concrete mixture (e.g., cementitious material, aggregate, sand, etc.), through the hopper **236**. In some embodiments, the mixer truck **200** includes an injection system (e.g., a series of nozzles, hoses, and/or valves). The injection system may include an injection valve that selectively fluidly couples a supply of fluid to the inner volume of the mixing drum **232**. In one embodiment, the injection system is used

to inject water and/or chemicals (e.g., air entrainers, water reducers, set retarders, set accelerators, superplasticizers, corrosion inhibitors, coloring, calcium chloride, minerals, and/or other concrete additives, etc.) into the mixing drum **232**. The injection valve may facilitate injecting water and/or chemicals from a fluid reservoir (e.g., a water tank, etc.) into the mixing drum **232**, while preventing the mixture in the mixing drum **232** from exiting the mixing drum **232** through the injection system. In some embodiments, one or more mixing elements (e.g., fins, etc.) are positioned in the interior of the mixing drum **232**, and may be configured to agitate the contents of the mixture when the mixing drum **232** is rotated in a first direction (e.g., counterclockwise, clockwise, etc.), and drive the mixture out through the chute **238** when the mixing drum **232** is rotated in a second direction (e.g., clockwise, counterclockwise, etc.). In some embodiments, the chute **238** includes an actuator positioned such that the chute **238** may be selectively pivotable to position the chute **238** (e.g., vertically, laterally, etc.), for example, at an angle at which the mixture is expelled from the mixing drum **232**.

(55) As shown in FIG. 5, the vehicle **10** is configured as a fire fighting vehicle or fire apparatus (e.g., a turntable ladder truck, a pumper truck, a quint, etc.), shown as fire fighting vehicle **300**. As shown in FIG. 5, the fire fighting vehicle **300** is configured as a rear-mount aerial ladder truck. In other embodiments, the fire fighting vehicle **300** is configured as a mid-mount aerial ladder truck, a quint fire truck (e.g., including an on-board water storage, a hose storage, a water pump, etc.), a tiller fire truck, a pumper truck (e.g., without an aerial ladder), or another type of response vehicle. According to an exemplary embodiment, the vehicle **10** is configured as a police vehicle, an ambulance, a tow truck, or still other vehicles used for responding to a scene (e.g., an accident, a fire, an incident, etc.).

(56) As shown in FIG. 5, in the fire fighting vehicle **300**, the application kit **80** is positioned mainly rearward from the cab **40**. The application kit **80** includes deployable stabilizers (e.g., outriggers, downriggers, etc.), shown as outriggers **330**, that are coupled to the chassis **20**. The outriggers **330** may be configured to selectively extend from each lateral side and/or the rear of the fire fighting vehicle **300** and engage a support surface (e.g., the ground) in order to provide increased stability while the fire fighting vehicle **300** is stationary. This increased stability is desirable when the ladder assembly **308** is in use (e.g., extended from the fire fighting vehicle **300**) to prevent tipping. In some embodiments, the application kit **80** further includes various storage compartments (e.g., cabinets, lockers, etc.) that are selectively opened and/or accessed for storage and/or component inspection, maintenance, and/or replacement.

(57) As shown in FIG. 5, the application kit **80** includes a ladder assembly **308** coupled to the chassis **20**. The ladder assembly **308** includes a series of ladder sections **340** that are slidably coupled with one another such that the ladder sections **340** may extend and/or retract (e.g., telescope) relative to one another to selectively vary a length of the ladder assembly **308**. A base platform, shown as turntable **342**, is rotatably coupled to the chassis **20** and to a proximal end of a base ladder section **340** (i.e., the most proximal of the ladder sections **340**). The turntable **342** may be configured to rotate about a vertical axis relative to the chassis **20** to rotate the ladder sections **340** about the vertical axis (e.g., up to 360 degrees, etc.). The ladder sections **340** may rotate relative to the turntable **342** about a substantially horizontal axis to selectively raise and lower the ladder sections **340** relative to the chassis **20**. As shown, a water turret or implement, shown as monitor **344**, is coupled to a distal end of a fly ladder section **340** (i.e., the most distal of the ladder sections **340**). The monitor **344** may be configured to expel water and/or a fire suppressing agent (e.g., foam, etc.) from a water storage tank and/or an agent tank onboard the fire fighting vehicle **300**, and/or from an external source (e.g., a fire hydrant, a separate water/pumper truck, etc.). In some embodiments, the ladder assembly **308** further includes an aerial platform coupled to the distal end of the fly ladder section **340** and configured to support one or more operators.

(58) According to another exemplary embodiment, as shown in FIG. 6, the vehicle **10** is configured as a fire fighting vehicle, shown as airport rescue and fire fighting (ARFF) truck **400**. As shown in

FIG. 6, the application kit **80** is positioned primarily rearward of the cab **40**. As shown, the application kit **80** includes a series of storage compartments or cabinets, shown as compartments **430**, that are coupled to the chassis **20**. The compartments **430** may store various equipment or components of the ARFF truck **400**.

(59) The application kit **80**, as shown in FIG. 6, includes a pump system **432** (e.g., an ultra-high-pressure pump system, etc.) positioned within one of the compartments **430** near the center of the ARFF truck **400**. The application kit **80** further includes a water tank **434**, an agent tank **436**, and an implement or water turret, shown as monitor **438**. The pump system **432** may include a high pressure pump and/or a low pressure pump, which may be fluidly coupled to the water tank **434** and/or the agent tank **436**. The pump system **432** may to pump water and/or fire suppressing agent from the water tank **434** and the agent tank **436**, respectively, to the monitor **438**. The monitor **438** may be selectively reoriented by an operator to adjust a direction of a stream of water and/or agent. As shown in FIG. 6, the monitor **438** is coupled to a front end of the cab **40**.

(60) As shown in FIG. 7, the vehicle **10** is configured as a lift device, shown as boom lift **500**. The boom lift **500** may be configured to support and elevate one or more operators. In other embodiments, the vehicle **10** is configured as another type of lift device that is configured to lift operators and/or material, such as a skid-loader, a telehandler, a scissor lift, a fork lift, a vertical lift, and/or any other type of lift device or machine.

(61) As shown in FIG. 7, the application kit **80** includes a base assembly, shown as turntable **504** that is rotatably coupled to the chassis **20**. The turntable **504** may be configured to selectively rotate relative to the chassis **20** about a substantially vertical axis. In some embodiments, the turntable **504** includes a counterweight positioned near the rear of the turntable **504**. The turntable **504** is rotatably coupled to a lift assembly, shown as boom assembly **508**. The boom assembly **508** includes a first section or telescoping boom section, shown as lower boom **540**. The lower boom **540** includes a series of nested boom sections that extend and retract (e.g., telescope) relative to one another to vary a length of the boom assembly **508**. The boom assembly **508** further includes a second boom section or four bar linkage, shown as upper boom **542**. The upper boom **542** may include structural members that rotate relative to one another to raise and lower a distal end of the boom assembly **508**. In other embodiments, the boom assembly **508** includes more or fewer boom sections (e.g., one, three, five, etc.) and/or a different arrangement of boom sections.

(62) As shown in FIG. 7, the boom assembly **508** includes a first actuator, shown as lower lift cylinder **544**. The lower boom **540** is pivotally coupled (e.g., pinned, etc.) to the turntable **504** at a joint or lower boom pivot point. The lower lift cylinder **544** (e.g., a pneumatic cylinder, an electric actuator, a hydraulic cylinder, etc.) is coupled to the turntable **504** at a first end and coupled to the lower boom **540** at a second end. The lower lift cylinder **544** may be configured to raise and lower the lower boom **540** relative to the turntable **504** about the lower boom pivot point.

(63) The boom assembly **508** further includes a second actuator, shown as upper lift cylinder **546**. The upper boom **542** is pivotally coupled (e.g., pinned) to the upper end of the lower boom **540** at a joint or upper boom pivot point. The upper lift cylinder **546** (e.g., a pneumatic cylinder, an electric actuator, a hydraulic cylinder, etc.) is coupled to the upper boom **542**. The upper lift cylinder **546** may be configured to extend and retract to actuate (e.g., lift, rotate, elevate, etc.) the upper boom **542**, thereby raising and lowering a distal end of the upper boom **542**.

(64) As shown in FIG. 7, the application kit **80** further includes an operator platform, shown as platform assembly **550**, coupled to the distal end of the upper boom **542** by an extension arm, shown as jib arm **552**. The jib arm **552** may be configured to pivot the platform assembly **550** about a lateral axis (e.g., to move the platform assembly **550** up and down, etc.) and/or about a vertical axis (e.g., to move the platform assembly **550** left and right, etc.).

(65) According to an exemplary embodiment, the platform assembly **550** provides a platform configured to support one or more operators or users. In some embodiments, the platform assembly **550** includes accessories or tools configured for use by the operators. In one embodiment, the

platform assembly **550** includes pneumatic tools (e.g., an impact wrench, airbrush, nail gun, ratchet, etc.), plasma cutters, welders, spotlights, etc. In other embodiments, the platform assembly **550** includes a control panel (e.g., a user interface, a removable or detachable control panel, etc.) configured to control operation of the boom lift **500** (e.g., the turntable **504**, the boom assembly **508**, etc.) from the platform assembly **550** or remotely. In other embodiments, the platform assembly **550** is omitted, and the boom lift **500** includes an accessory and/or tool (e.g., forklift forks, etc.) coupled to the distal end of the boom assembly **508**.

(66) According to an exemplary embodiment, as shown in FIG. **8**, the vehicle **10** is configured as a lift device, shown as scissor lift **600**. As shown in FIG. **8**, the application kit **80** includes a body, shown as lift base **604**, coupled to the chassis **20**. The lift base **604** is coupled to a scissor assembly, shown as lift assembly **608**, such that the lift base **604** supports the lift assembly **608**. The lift assembly **608** is configured to extend and retract, raising and lowering between a raised position and a lowered position relative to the lift base **604**.

(67) As shown in FIG. **8**, the lift base **604** includes a series of actuators, stabilizers, downriggers, or outriggers, shown as leveling actuators **630**. The leveling actuators **630** may extend and retract vertically between a stored position and a deployed position. In the stored position, the leveling actuators **630** may be raised, such that the leveling actuators **630** do not contact the ground. Conversely, in the deployed position, the leveling actuators **630** may engage the ground to lift the base assembly **604**. The length of each of the leveling actuators **630** in their respective deployed positions may be varied in order to adjust the pitch (e.g., rotational position about a lateral axis) and the roll (e.g., rotational position about a longitudinal axis) of the base assembly **604** and/or the chassis **20**. Accordingly, the lengths of the leveling actuators **630** in their respective deployed positions may be adjusted to level the base assembly **604** with respect to the direction of gravity (e.g., on uneven, sloped, pitted, etc. terrain). The leveling actuators **630** may lift the wheel and tire assemblies **54** off of the ground to prevent movement of the scissor lift **600** during operation. In other embodiments, the leveling actuators **630** are omitted.

(68) According to an exemplary embodiment, the lift assembly **608** includes a series of subassemblies, shown as scissor layers **640**, each including a pair of inner members **642** and a pair of outer members **644**. The scissor layers **640** may be stacked atop one another in order to form the lift assembly **608**. The inner members **642** may be pivotally coupled to the outer members **644** near the center of both the inner members **642** and the outer members **644**. In this regard, the inner members **642** may pivot relative to the outer members **644** about a lateral axis. Each of the inner members **642** and the outer members **644** may include a top end and a bottom end. The bottom end of each inner member **642** may be pivotally coupled to the top end of the outer member **644** immediately below it, and the bottom end of each outer member **644** may be pivotally coupled to the top end of the inner member immediately below it. Accordingly, each of the scissor layers **640** may be coupled to one another such that movement of one scissor layer **640** causes a similar movement in all of the other scissor layers **640**. The bottom ends of the inner member **642** and the outer member **644** that make up the lowermost scissor layer **640** may be coupled to the base assembly **604**. The top ends of the inner member **642** and the outer member **644** that make up the uppermost scissor layer **640** may be coupled to the platform assembly **650**. In some embodiments, scissor layers **640** may be added to, or removed from, the lift assembly **608** in order to increase, or decrease, the fully extended height of the lift assembly **608**.

(69) As shown in FIG. **8**, the lift assembly **608** also includes one or more lift actuators **646** (e.g., hydraulic cylinders, pneumatic cylinders, motor-driven leadscrews, etc.) configured to extend and retract the lift assembly **608**. The lift actuators **646** may be pivotally coupled to an inner member **642** at a first end and pivotally coupled to an inner member **642** of another scissor layer **640** at a second end. In an exemplary embodiment, these inner members **642** belong to a first scissor layer **640** and a second scissor layer **640** (which may be separated by a third scissor layer **640**). In other embodiments, the lift actuators **646** are arranged in other configurations (e.g., the first scissor layer

640 and the second scissor layer **640** are not separated by a third scissor layer **640**, etc.).

(70) According to an exemplary embodiment, as distal or upper end of the lift assembly **608** is coupled to an operator platform, shown as platform assembly **650**. The lift actuators **646** may be configured to actuate the lift assembly **608** to selectively reposition the platform assembly **650** between a lowered position (e.g., where the platform assembly **650** is proximate to the lift base **604**) and a raised position (e.g., where the platform assembly **650** is at an elevated height relative to the lift base **604**). Specifically, in some embodiments, extension of the lift actuators **646** moves the platform assembly **650** upward (e.g., extending the lift assembly **608**), and retraction of the lift actuators **646** moves the platform assembly **650** downward (e.g., retracting the lift assembly **608**). In other embodiments, extension of the lift actuators **646** retracts the lift assembly **608**, and retraction of the lift actuators **646** extends the lift assembly **608**. In some embodiments, the outer members **644** are parallel to and/or in contact with one another when the lift assembly **608** is in the stored position.

(71) In some embodiments, the platform assembly **650** includes a platform that is configured to support one or more operators or users. Similar to the platform assembly **550**, the platform assembly **650** may include accessories or tools (e.g., pneumatic tools, plasma cutters, welders, spotlights, etc.) configured for use by an operator. The platform assembly **650** may include a control panel to control operation of the scissor lift **600**.

Battery Box

(72) According to an exemplary embodiment, as shown in FIGS. **9A** and **9B**, a vehicle **10**, includes a chassis **20**. The vehicle **10** may be any vocational vehicle. The chassis **20** may extend along the length of the vehicle **10**. The chassis **20** may include at least one of a structural housing, shown as battery box or enclosure **1002**, and a subframe **1004**, and any combination thereof. In one exemplary embodiment, the chassis **20** is one contiguous battery box **1002**. The battery box **1002** may be made of any material strong enough to support the various components and subcomponents of the vehicle **10** and to protect the batteries **60** and other systems contained therein. The battery box **1002** may include integrated suspension hard points such that it acts as a typical frame or chassis of a vehicle by providing the necessary control and stability to the vehicle **10**. According to an exemplary embodiment, the battery box **1002** may include sufficient springs, shock absorbers, and linkages that connect the wheels **54** to the vehicle **10** and allow relative motion between the vehicle **10** and the wheels **54**. The one contiguous battery box **1002** may extend from a front end **1006** of the vehicle **10** (e.g., an end proximate the cab **40**) to a rear end **1008** of the vehicle **10** (e.g., an end proximate the body **80**). An internal cavity of the battery box **1002** may be one large open space. In another embodiment, the battery box **1002** is divided into compartments. In another exemplary embodiment, the chassis **20** includes the battery box **1002** and at least one subframe **1004**. The subframe **1004** may provide support to the portions of the vehicle **10** that are not supported by the battery box **1002**. The battery box **1002** may be coupled with the subframe **1004**. In some embodiments, the chassis **20** includes at least one of a front subframe **1004** and a rear subframe **1004**. If the volume of the battery box **1002** disposed between front axle **50** and rear axles **52** of the vehicle **10** provides enough space for batteries **60** and other systems, then the remainder of the chassis **20** extending over the axles toward the front end **1006** and the rear end **1008** of the vehicle **10** can include a subframe **1004**.

(73) According to an exemplary embodiment, the battery box **1002** has at least one depth **1010**. In some embodiments, the depth **1010** is measured along a direction that is perpendicular to a ground on which the vehicle **10** travels. The depth **1010** may be based, in part, on the size of contents (e.g., systems, batteries, modules) disposed in the battery box **1002**. In one embodiment, if the entire battery box **1002** is disposed between the front axle **50** and the rear axles **52** of the vehicle **10** (e.g., the chassis **20** includes a rear subframe **1004** and a front subframe **1004**), the battery box **1002** has one uniform depth **1010**. In another embodiment, if the battery box **1002** extends toward the rear end **1008** of the vehicle **10** over the rear axles **52**, a first portion **1022** disposed between the front

and rear axles **50**, **52** has a first depth **1010** and a second portion **1024** disposed above the rear axles **52** has a second depth **1010**. According to an exemplary embodiment, the first depth **1010** is greater than the second depth **1010**. Any portion of the battery box **1002** may include more than one depth **1010**. In another embodiment, the battery box **1002** has a lattice structure. The lattice structure reduces weight by removing material from the battery box **1002** that is unnecessary, provides means for positioning and securing contents disposed within the battery box **1002**, and provides beneficial structural properties (e.g., strength, flexibility).

(74) According to an exemplary embodiment, as shown in FIG. **9B**, the front subframe **1004** may be coupled with a front side **1012** of the battery box **1002**. In some embodiments, the front side **1012** of the battery box **1002** may be disposed at a position directly adjacent the wheels **54** coupled with a front axle **50**. In other words, the front side **1012** of the battery box **1002** may define the same longitudinal position as the front axle **50** (e.g., the front side **1012** may align with an central axis defined along the front axle **50**). In some embodiments, the front side **1012** of the battery box **1002** may be arranged anywhere between the wheels **54** coupled with the front axle **50** and the wheels **54** coupled with a rear axle **52**. In some embodiments, the front side **1012** of the battery box **1002** is disposed halfway between the front axle **50** and the front-most rear axle **52**. The front subframe **1004** may extend from the front end **1006** of the vehicle **10** (or beyond the front end **1006**) to the position where the front side **1012** of the battery box **1002** is disposed. The chassis **20** may also be configured such that a subframe **1004** can be coupled with, either directly or indirectly, a top, bottom, or side of the battery box **1002**. In another embodiment, the battery box **1002** can have an external connection point (e.g., a bracket) disposed on the bottom of the battery box **1002**. The external connection point can be configured to couple with the subframe **1004**.

(75) In some embodiments, the battery box **1002** extends away from the front side **1012** of the battery box **1002** toward the rear end **1008** of the vehicle **10**. In other embodiments, a back side **1014** of the battery box **1002** is disposed in front of the wheels **54** coupled with the front-most rear axle **52**. In other embodiments, the entire battery box **1002** is disposed between the front axle **50** and the front-most rear axle **52**. In such an embodiment, the battery box **1002** has a width **1016** that extends a full width of the vehicle **10** (e.g., the width **1016** of the battery box **1002** is approximately equal to a lateral distance between the laterally-outwardmost points defined by the wheels **54**). As such, a right side **1018** of the battery box **1002** aligns with a right side of the vehicle **10** and a left side **1020** of the battery box **1002** aligns with a left side **1020** of the vehicle **10** (e.g., when looking from the rear end **1008** toward the front end **1006**). The width **1016** may also be larger than the width of the vehicle **10** or smaller than the width of the vehicle **10**. The battery box **1002** may also have more than one width **1016**. In another embodiment, the back side **1014** of the battery box **1002** is disposed at or around the rear end **1008** of the vehicle **10**. In such an embodiment, a first portion **1022** of the battery box **1002** adjacent to the front side **1012** has a first width **1016** and a second portion **1024** of the battery box **1002** adjacent to the rear end **1008** has a second width **1016**, with the second width **1016** being less than the first width **1016**. As such, the first portion **1022** may be disposed between the front axle **50** and the rear axles **52**. The first portion **1022** may have a width **1016** proximate the width of the vehicle **10**. The second portion **1024** may be disposed above the rear axles **52** and between the wheels **54** disposed on the left side of the vehicle **10** and the wheels **54** disposed on the right side of the vehicle **10**. To fit between the wheels **54**, the width **1016** of the second portion **1024** may be smaller than the width **1016** of the first portion **1022**. In some embodiments, as shown in FIG. **11A**, among others, the battery box **1002** may have a third portion **1026** that extends toward the front end **1006** from the first portion **1022**. The third portion **1026** may be disposed above the front axle **50** and between the front wheels **54** of the vehicle **10**. The third portion **1026** has a third width **1016**. The third width **1016** may be the same as or different from the first or second width **1016**.

(76) According to an exemplary embodiment, the back side **1014** of the battery box **1002** is disposed anywhere between the front side **1012** of the battery box **1002** and the rear end **1008** of

the vehicle **10** (or beyond the rear end **1008**). As shown in FIG. **9B**, the back side **1014** of the battery box **1002** extends beyond (e.g., longitudinally over and past) the rear-most rear axle **52** and is disposed at a position proximate the rear end **1008** of the vehicle **10**. In another embodiment, the back side **1014** of the battery box **1002** is disposed at a position in front of the front-most rear axle **52**. In such an embodiment, the chassis **20** includes a rear subframe **1004** extending at least from the back side **1014** of the battery box **1002** to the rear end **1008** of the vehicle **10**. The rear subframe **1004** may be coupled with the battery box **1002** in a similar manner as the front subframe **1004** as described herein.

(77) The battery box **1002** may comprise any shape or orientation. As shown in FIG. **9B**, the first portion **1022** and the second portion **1024** of the battery box **1002** have a rectangular shape with square corners. A center line of the first portion **1022** may align with a center line of the second portion **1024**. The first portion **1022** may transition to the second portion **1024** at a forty-five degree angle. In other embodiments, the first portion **1022** is oval and the second portion **1024** is square with rounded corners. The first portion **1022** may offset from the second portion **1024** (e.g., the centerlines do not align). The first portion **1024** may transition to the second portion **1022** with differing angles.

(78) According to an exemplary embodiment, as shown in FIG. **10**, the battery box **1002** is configured to accommodate a tag (or auxiliary) axle **1052**. The battery box **1002** may be reinforced (e.g., add structural bracing, use stronger material) at a location where the auxiliary axle **1052** can be coupled with the battery box **1002** (e.g., bolted to a side of the battery box **1002**). According to an exemplary embodiment, the width **1016** of the battery box **1002** is such that the wheel **54** coupled with the auxiliary axle **1052** does not extend beyond the width of the vehicle **10**. The battery box **1002** may be configured to allow the auxiliary axle **1052** to be added or removed from the vehicle **10** at any time.

(79) According to an exemplary embodiment, as shown in FIGS. **11A** and **11B**, the battery box **1002** includes a shell **1102** that defines an internal cavity **1104**. The internal cavity **1104** may be configured to receive at least one module **1106**. The module **1106** may be or include at least one battery **60**. The module **1106** may include at least one gripping component **1108** configured to assist in installation and removal of the module **1106** from the internal cavity **1104** of the battery box **1002**. In some embodiments, the gripping component **1108** is a handle. The handle may be any shape, size, or material. The handle may be coupled with the module **1106** at any location such that a person or a machine can grip the handle when installing or removing the module **1106**. In another embodiment, the gripping component **1108** is a magnet such that a person or a machine can gain control of the module **1106** by engaging the magnet. In another embodiment, the gripping component **1108** is a machinery pocket (e.g., a forklift pocket) such that a machine can control the placement and orientation of the module **1106** such that the machine can perform at least one of inserting the module **1106** into the internal cavity **1104** of the battery box **1002** and removing the module **1106** from the internal cavity **1104** of the battery box **1002**.

(80) The module **1106** may be configured to provide energy to the vehicle **10** to facilitate the operation of the components of the vehicle **10**. In some embodiments, the vehicle **10** is refuse vehicle **100** including a lift assembly **108**. The energy from the module **1106** may be used to operate the lift assembly **108** of the refuse vehicle **100**. The battery box **1002** may be configured to transfer the energy from the module **1106** to the components of the vehicle **10**.

(81) The module **1106** may include a plurality of internal components and systems. According to an exemplary embodiment, as shown in FIG. **16**, the module **1106** includes an electrical energy storage device (e.g., battery **60**), associated cabling **1602**, a thermal management system **1604**, and a communication system **1606**. The battery **60** may be configured to provide power or energy to the module **1106** to be provided to a component or system. The associated cabling **1602** may connect the battery **60** to a module terminal **1110** or other component of the module **1106** such that energy can be transferred between components via the cabling **1602**. The thermal management system

1604 may control a temperature within the module **1106**. In some embodiments, the thermal management system **1604** includes at least one fan **1608** to cool the module **1106**. The communication system **1606** may facilitate communication between the module **1106** and external systems. In some embodiments, the communication system **1606** may transmit a signal to an external computing system indicating a state of the module **1106** (e.g., low power, faulty component, etc.) In some embodiments, the module **1106** includes other components and systems. In some embodiments, the module **1106** includes any combination of systems and components. The module **1106** may be configured to be assembled off the vehicle **10** and then installed as a single unit. According to an exemplary embodiment, the battery box **1002** is configured to include a single module **1106**. As such, if one component of the module **1106** needs replacing, repair, etc., the module **1106** and any components contained therein may be removed from the vehicle **10** by removing the single module **1106** from the internal cavity **1104** of the battery box **1002**. In another embodiment, as shown in FIG. **11A**, the internal cavity **1104** of the battery box **1002** is configured to receive a plurality of modules **1106**. Each module **1106** can include separate electrical energy storage devices, wiring, systems, etc. With a plurality of modules **1106**, if a component of one module **1106** needs to be charged, repaired, replaced, etc., the module **1106** may be removed from the battery box **1002** while the remaining modules **1106** can stay in the battery box **1002**. In some embodiments, the installation and removal of a first module **1106** is independent from the installation and removal of a second module **1106**. If the electrical energy storage devices in the first module **1106** run out of energy, the first module **1106** may be removed from the battery box **1002**. The electrical energy storage device in the first module **1106** may be charged or replaced such that the first module **1106** can be reinstalled in the battery box **1002**, or the first module **1106** can be replaced with a second module **1106** with an already-charged battery.

(82) According to an exemplary embodiment, the battery box **1102** is configured to receive supplemental energy when the energy of a module **1106** is low or depleted. In some embodiments, supplemental energy includes recharging an electrical energy storage device. In other embodiments, supplemental energy includes replacing an electrical energy storage device, or replacing a module **1106**. According to an exemplary embodiment, the battery box **1002** is configured to recharge a battery **60** via an external electrical power source (e.g., a generator) when the battery **60** no longer has energy to provide to the components of a vehicle **10**. The battery box **1002** may also be configured to replace the battery **60** with a new, already-charged, battery **60** by removing the module **1106** from the battery box **1002**, removing a first battery **60**, and replacing the first battery **60** with a second battery **60**. The battery box **1002** may also be configured to replace an entire module **1106**. According to an exemplary embodiment, a first module **1106** is replaced with a second module **1106** with already-charged batteries **60** when multiple batteries **60** within the first module **1106** are low on energy.

(83) According to an exemplary embodiment, the shell **1102** of the battery box **1002** is configured to protect and regulate the components disposed in the battery box **1002**. The battery box **1002** may be configured to protect the module **1106** from external elements including, but not limited to, water, salt, dust, mud, and debris. In some embodiments, the shell **1102** creates an airtight seal around the internal cavity **1104** such that no external elements can come into contact with the module **1106** or any other components or systems disposed within the battery box **1002**. In some embodiments, the shell **1102** includes module protection **1028** that protect the modules **1106** within the battery box **1002**. The module protection **1028** may provide impact protection to the module **1106** and any other internal components. In some embodiments, the module protection **1028** may be a material, a layer of material, or other impact-resistant devices (e.g., springs) configured to lessen or dampen the impact of a force such that the module **1106** is not broken or damaged when undergoing external forces. In some embodiments, the shell **1102** includes module protection **1028** that thermally insulates the module **1106** and other internal components from outside ambient conditions. The shell **1102** may comprise a material, or a layer of material, that insulates the

internal cavity **1104** such that the temperature and humidity within the internal cavity **1104** are not affected by the temperature and humidity outside of the battery box **1002**. In some embodiments, the module protection **1028** is disposed along at least a portion of an inner wall of the shell.

(84) According to an exemplary embodiment, the battery box **1002** is configured such that a component of the vehicle **10** can operate when at least one module **1106** is missing from the battery box **1002** (e.g., part of the internal cavity **1104** of the battery box **1002** designated for a module **1106** is empty) or the batteries **60** in the module **1106** are out of energy. In some embodiments, a battery box **1002** includes a plurality of modules **1106**. The plurality of modules **1106** may include a first module **1106**, a second module **1106**, and a third module **1106**. If the first module **1106** is removed, or the batteries **60** within the first module **1106** run out of energy, the energy from the second and third modules **1106** may still be transferred to the components of the vehicle **10**. If that energy is enough to operate the components, then operation can continue.

(85) The operation of the components of the vehicle **10** may also depend on how the energy from the modules **1106** in the battery box **1002** is distributed. According to an exemplary embodiment, each module **1106** within a plurality of modules **1106** provides energy directly to a designated component. In some embodiments, a first module **1106** is configured to power a lift assembly **108** of a refuse vehicle **100** and a second module **1106** is configured to power a tailgate **132** of the same refuse vehicle **100**. As such, if the first module **1106** runs out of energy, the tailgate **132** can still operate, but the lift assembly **108** cannot.

(86) In some embodiments, as shown in FIG. **17**, the battery box **1002** includes a power distribution unit **1702** (e.g., a power bus). The power distribution unit may be configured to receive energy from the plurality of modules **1106** and selectively direct the energy to the components of the vehicle **10** to be operated. In some embodiments, a first module **1106** and a second module **1106** are electrically coupled with a power distribution unit **1702**. The first and second module **1106** may be electrically coupled with the power distribution unit **1702** via a cable **1704** (e.g., a wire), terminals (e.g., a module terminal **1110** and a system terminal **1112**), etc. The power distribution unit **1702** may be configured to receive energy from the first module **1106** and the second module **1106** and selectively direct the energy from the first module **1106** and the second module **1106** to at least one component of the vehicle **10** to be operated. If the first module **1106** (corresponding to the lift assembly **108**) has no remaining energy or is missing from the battery box **1002**, the energy from the second module **1106** may be used to operate the lift assembly **108**.

(87) According to an exemplary embodiment, the module **1106** includes at least one module terminal **1110**. The module terminal **1110** may be configured to facilitate the transfer of energy from the batteries **60** in the module **1106** to the components of the vehicle **10**. In some embodiments, the module terminal **1110** comprises a conductive material such that the energy from the battery **60** can be transferred through the module terminal **1110** to a receiving terminal. As such an embodiment, like a typical battery, the module terminal **1110** can have a positive terminal and a negative terminal configured to connect to a receiving terminal. The battery box **1002** may include at least one receiving terminal, shown as system terminal **1112**. The system terminal **1112** may be configured to contact a module terminal **1110** of a module **1106** to facilitate the transfer of energy from the module **1106** to a component of the vehicle **10**. The system terminal **1112** may comprise a conductive material such that the system terminal **1112** can be configured to receive the energy from the module **1106**. In some embodiments, the system terminal **1112** is a bus bar such that contact made with the module terminal **1110** can be made at different locations along the system terminal **1112**. To transfer the energy from the battery **60** to the components of the vehicle **10**, module **1106** may be configured such that the module terminal **1110** contacts the system terminal **1112** when properly positioned within the battery box **1002**. The module terminal **1110** may be any shape or size capable of sufficiently contacting the system terminal **1112** to facilitate the transfer of energy. Similarly, the system terminal **1112** may be any shape or size. In one exemplary embodiment, the battery box **1002** includes a single system terminal **1112** configured to contact the

module terminal **1110** of each module **1106** (e.g., one large system terminal **1112** extending the longitudinal length of the battery box **1002** contacting a plurality of module terminals **1110**). In another embodiment, the battery box **1002** includes a plurality of system terminals **1112** wherein each system terminal **1112** corresponds to an individual module terminal **1110** (e.g., a first system terminal **1112** contacts a first module terminal **1110**). In another embodiment, a single system terminal **1112** corresponds to a subset of a plurality of module terminals **1110** (e.g., a first system terminal **1112** contacts a first subset of a plurality of module terminals **1110** and a second system terminal **1112** contacts a second subset of the plurality of module terminals **1110**). The system terminal **1112** may be configured such that a module terminal **1110** can contact the system terminal **1112** at different locations. According to an exemplary embodiment, a first module **1106** has a module terminal **1110** that contacts a corresponding system terminal **1112** at a center of the system terminal **1112**. In other embodiments, the first module **1106** is replaced with a second module **1106** and the second module **1106** has a module terminal **1110** that contacts the corresponding system terminal **1112** at a location offset from the center of the system terminal **1112**.

(88) In another exemplary embodiment, a system terminal **1112** is configured to receive energy from a plurality of modules **1106** while only contacting one module terminal **1110**. In some embodiments, modules **1106** are connected in series such that energy flows from a first module **1106** to a second module **1106**. The first module **1106** may include a module terminal **1110**. The second module **1106** may include two module terminals **1110**. A receiving module terminal **1110** of the second module **1106** may contact the module terminal **1110** of the first module **1106**. A supplying module terminal **1110** of the second module **1106** may contact either a module terminal **1110** of a third module **1106** or a system terminal **1112** of the battery box **1002**. The energy may be transferred between any number of modules **1106** before being transferred to the vehicle **10** via the system terminal **1112**.

(89) According to an exemplary embodiment, as shown in FIG. **18**, the battery box **1002** includes a thermal management system **1802**. The thermal management system **1802** may include various components to control a temperature within the internal cavity **1104** of the battery box. In some embodiments, the thermal management system **1802** includes at least one of a fan **1804**, a temperature sensor **1806**, a vent **1808**, and a heating element **1810**. The thermal management system **1802** may include a controller **1812**. The controller **1812** may have a processor **1814** and a memory **1816** to facilitate actuation of the various components of the thermal management system **1802**. For example, the memory **1816** may store instructions that include thresholds that dictate when certain components are to be actuated. The processor **1814** may receive and analyze various inputs and determine which component to actuate. For example, the processors **1814** may receive a signal from temperature sensor **1806** indicating a temperature above a threshold stored in the memory **1816**, and send a signal to actuate the fan **1804**. In some embodiments, heat from systems disposed within the battery box **1002**, from the environment external to the battery box **1002**, etc. may cause a temperature of the internal cavity **1104** of the battery box **1002** to increase to a temperature that can be problematic or harmful to the components within the battery box **1002**. In some embodiments, the thermal management system **1802** includes a temperature sensor **1806** disposed within the internal cavity **1104** of the battery box **1002**. The temperature sensor **1806** may be configured to detect the temperature and cause a components of the thermal management system **1802** to activate to counteract the increase in temperature. In some embodiments, a detection of a temperature above ninety degrees causes the controller **1812** to actuate a fan **1804** disposed in the internal cavity **1104** of the battery box **1002** to blow air over the modules **1106** and other systems disposed in the battery box **1002** to cool the components down. In some embodiments, detecting the temperature causes the controller **1812** to open a vent **1808** such that the hot air can escape from the battery box **1002**. The thermal management system **1802** may also increase the temperature within the internal cavity **1104** of the battery box **1002** when the surrounding temperature is so low it negatively affects the internal components of the battery box **102**.

According to an exemplary embodiment, a temperature sensor detects the low temperature within the internal cavity **1104** and the controller **1812** of the thermal management system **1802** actuates a heating element **1810** (e.g., a resistive heater, an electric heater, a heat exchanger, etc.) to provide heat to the internal cavity **1104**. The thermal management system **1802** may include any variety of components configured to regulate the temperature within the internal cavity **1104** of the battery box **1002**.

(90) According to an exemplary embodiment, the battery box **1002** includes at least one mounting provision **1114**. The mounting provision **1114** may be configured to facilitate the positioning of a module **1106** within the internal cavity **1104** of the battery box **1002** such that energy from the module **1106** can be transferred to the components of the vehicle **10**. In some embodiments, the mounting provision **1114** facilitates the proper placement of the module **1106** such that the module terminal **1110** of the module **1106** contacts the system terminal **1112** of the battery box **1002**. In some embodiments, the mounting provision **1114** is a guide configured to direct a module **1106** toward a desired location within the battery box **1002**. The guide may be a wall or plate configured to create at least one compartment **1126** within the battery box **1002**. Each wall may be arranged to create a compartment **1126** with a shape and size to accommodate a module **1106**. In some embodiments, a plurality of walls extend laterally from a left side **1020** of the battery box **1002** to a right side **1018** of the battery box **1002** such that a module **1106** can slide between the walls. A compartment **1126** may be defined by the space between at least two walls or between a wall and the shell **1102** of the battery box **1002**. The sizes of the compartments **1126** may vary depending, in part, on the size of the module **1106** it is intended to accommodate. Modules **1106**, and therefore the compartments **1126**, may have any shape or size.

(91) In some embodiments, the size of the module **1106** may be based, at least in part, on the size of the battery box **1002**. For example, as shown in FIG. **11A**, the first portion **1022** of the battery box **1002** has a greater width **1016** than the second portion **1024** and the third portion **1026**. A module **1106** disposed in the first portion **1022** may have a first module width **1128** and a module **1106** disposed in the second portion **1024** may have a second module width **1128**. The first module width may be greater than the second module width **1128**. The module **1106** may extend the full width **1016** of the battery box **1002** or just part of the width **1016** such that the module width **1128** may be less than or equal to the width **1016** of the battery box **1002**. As shown in FIG. **11B**, the first portion **1022** of the battery box **1002** has a greater depth **1010** than the second portion **1024** and the third portion **1026**. A module **1106** disposed in the first portion **1022** may have a first module height **1130** and a module **1106** disposed in the second portion **1024** may have a second module height **1130**. The first module height **1130** may be greater than the second module height **1130**. The module **1106** may extend the full depth **1010** of the battery box **1002** or just part of the depth **1010** such that the module height **1130** may be less than or equal to the depth **1010** of the battery box **1002**.

(92) According to an exemplary embodiment, a battery box **1002** comprising a wall as a mounting provision **1114** has the system terminal **1112** disposed on the wall. The module terminal **1110** may be disposed on a side of the module **1106** that opposes the wall such that the module terminal **1110** contacts the system terminal **1112** when the module **1106** slides along the wall. The system terminal **1112** may also be disposed at different locations within the battery box **1002**. In some embodiments, the system terminal **1112** is on any side of the internal cavity **1104** of the battery box **1002** (e.g., top, bottom, left side, right side, front side, back side). A location of the module terminal **1110** on the module **1106** may depend, in part, on a location of the system terminal **1112** in the battery box **1002**. In some embodiments, the module terminal **1110** is on top of the module **1106** when the system terminal **1112** is on the top of the internal cavity **1104** of the battery box **1002**. In some embodiments, the module **1106** can be reoriented depending on the location of the system terminal **1112**. In such an embodiment, a single module **1106** can be used with system terminals **1112** at any location within the battery box **1002**. The module **1106** may be reoriented

(e.g., turned, flipped) such that the module terminal **1110** is facing an appropriate direction to contact the system terminal **1112**. In one embodiment, a first side of a module **1106** includes a module terminal **1110**. A top of the internal cavity **1104** includes a system terminal **1112**. The first side of the module **1106** may be directed toward the top of the internal cavity **1104** such that the module terminal **1110** can contact the system terminal **1112**. If the module **1106** is disposed in a different portion of the battery box **1002**, or in a different battery box **1002** entirely, and the system module **1112** is on a bottom of the internal cavity **1104**, the module **1106** may be flipped such that the first side is directed toward the bottom of the internal cavity **1104**.

(93) According to another exemplary embodiment, as shown in FIG. **12A**, the mounting provision **1114** of the battery box **1002** is a track or rail. Similar to the wall in FIG. **11A**, the track may guide the placement of the module **1106** such that the energy from the module **1106** can be transferred to the components of the vehicle **10** (e.g., the module terminal **1110** contacts the system terminal **1112**). In an exemplary embodiment, the system terminal **1112** is disposed on the top of the internal cavity **1104** of the battery box **1002** with a corresponding module terminal **1110** being disposed on the top of a module **1106**. Any mounting provision **1114** may be positioned directly next to another guide, spaced apart, extend in different direction, or have any other varied configuration. In some embodiments, a battery box **1002** includes a variety of mounting provisions **1114** (e.g., including both tracks and walls, among others).

(94) In some embodiments, the battery box **1002** includes other mounting provisions **1114** to facilitate proper placement and securing of the modules **1106**. The mounting provision **1114** may be any structural element configured to limit the movement of the module **1106** and keep the module **1106** at a specific location within the battery box **1002**. As shown in FIG. **12A**, the mounting provision **1114** is a movable member disposed at an end of a module **1106**. The movable member may be configured to move between a loading position (e.g., flush with the bottom of the internal cavity **1104** of the battery box **1002**) and a locking position (e.g., extending above the bottom of the internal cavity **1104** of the battery box **1002**). In such an embodiment, when installing a module **1106** into a battery box **1002**, an installer may push the movable member down into the loading position such that the path to slide the module **1106** into the battery box **1002** is unobscured. Once the module **1106** is far enough into the battery box **1002** such that the end of the module **1106** passes the location of the movable member, the movable member may bias back up into the locking position. To remove the module **1106**, the installer may push the movable member down again and slide the module **1106** out of the battery box **1002** over the location of the movable member. Once the module **1106** is no longer above the movable member, the movable member may be biased up into the locking position again. The movement of the movable member, in either direction, can be done manually or by a biasing force (e.g., a spring).

(95) According to another exemplary embodiment, a mounting provision **1114** includes a retractable pin and a corresponding receiver. A module **1106** may include a retractable pin on the bottom of the module **1106**. The pin may be biased to extend downward from the bottom of the module **1106**, and may be retracted into the module **1106** when force is applied (e.g., being pushed, connected to a button/lever on the module). In some embodiments, as a module **1106** moves along the bottom of the internal cavity **1104** of the battery box **1002**, the force from the bottom of the internal cavity **1104** forces the pin to the retracted position. The pin may correspond to a receiver (e.g., an aperture or hole) in the battery box **1002** such that when the pin aligns with the receiver, the pin biases to its extended position. When the pin is in the receiver, the module **1106** may no longer move until the pin is retracted. A module **1106** may include any number, any combination, and any arrangement of mounting provisions **1114** to facilitate the proper placement of the module **1106** in the battery box **1002**. Other mounting provisions **1114** may include various fasteners (e.g., magnets, bolts, screws, loops and hooks, straps) and other structures defining a location where a module **1106** can be positioned.

(96) According to an exemplary embodiment, the battery box **1002** includes integrated impact

protection for all enclosed systems within the battery box **1002**. The battery box **1002** may include cushioning material, springs, etc. configured to protect the enclosed systems (e.g., the module **1106**, wiring, power distribution unit) from damage from the vehicle **10** bouncing or vibrating due to rough terrain or from the vehicle **10** getting hit from other external forces (e.g., getting into a car crash). The integrated impact protection may be incorporated on one, all, or some of the sides of the internal cavity **1104** of the battery box **1002**. The impact protection may be incorporated between systems and components disposed within the battery box **1002**.

(97) According to an exemplary embodiment, as shown in FIG. **11B**, which is a side view of the battery box **1002** of FIG. **11A** the battery box **1002** includes mounting provisions **1114** in the form of walls. The system terminals **1112** are be coupled with the walls **1114**. The module terminals **1110** are coupled with a side of the modules **1106**. In some embodiments, the battery box **1002** is configured to include various sizes of modules **1106** and organize the modules **1106** in various arrangements. In some exemplary embodiments, a first module **1106** is stacked on top of a second module **1106** within the same compartment **1126** (e.g., between two adjacent walls **1114**). The first module **1106** may be supported by the second module **1106**. In another embodiment, the first module **1106** is separated from the second module **1106** by another structural element (e.g., a shelf). In another embodiment, some modules **1106** extend from the bottom of the internal cavity **1104** of the battery box **1002** to the top of the internal cavity **1104**, while others extend only part way, or others extend starting at an elevation above the bottom of the internal cavity **1104**.

(98) According to another exemplary embodiment, as shown in FIG. **12B**, the battery box **1002** includes a combination of mounting provisions **1114** (e.g., a track and a wall). The system terminals **1112** are coupled with a top of the internal cavity **1104** of the battery box **1002** or an internal shelf if a first module **1106** is to be stacked on top of a second module **1106**. The module terminals **1110** are coupled with a top of the modules **1106**. In another embodiment, when the first module **1106** is supported only by the second module **1106**, the first module **1106** includes two module terminals **1110**. A receiving module terminal **1110** may be on a bottom of the first module **1106** and a supplying module terminal **1110** may be on a top of the first module **1106**. The battery box **1002** may also include other mounting provisions **1114**, such as the movable member configured to move from a loading position to a locking position, among others.

(99) According to an exemplary embodiment, the battery box **1002** includes system routing pathways **1210** to route wire **1212** between appropriate components in order to transfer the energy from the modules **1106** to the appropriate location (e.g., component of the vehicle, power distribution unit). In some embodiments, the battery box **1002** includes conduits within the battery box **1002** connecting different systems within the battery box **1002** to each other. As shown in FIG. **12A**, a wall is the system routing pathway that contains the wiring **1212** and routes the wiring **1212** to a desired location. In another embodiment, as show in FIG. **12B**, the system routing pathway **1210** is a separate compartment where any wiring **1212** needed to connect the system terminals **1112** to a destination can be contained. The compartment may disposed on top of the battery box **1002** where wiring **1212** may be disposed in order to direct the energy from the modules **1106** to either a power distribution unit or directly to a component of the vehicle **10**. The power distribution unit may be contained within the battery box **1002**, or it can be external to the battery box **1002**.

(100) According to an exemplary embodiment, as shown in FIGS. **13A-13C**, the battery box **1002** includes at least one sealing mechanism **1120**. The sealing mechanism **1120** may be configured to allow or prevent installation and removal of modules **1106**. The sealing mechanism **1120** may be configured to protect the internal cavity **1104** of the battery box **1002** and any contents therein (e.g., modules **1106**) from water, salt, dust, mud, debris, among others. In some embodiments, the sealing mechanism **1120** includes a door configured to swing open and closed. In other embodiments, the sealing mechanism **1120** includes a panel configured to be secured and removed. In yet other embodiments, the sealing mechanism **1120** includes a locking mechanism configured to create and release airtight seals. As shown in FIG. **13A**, the battery box **1002** includes a door configured to

open and close when a module **1106** is to be installed or removed from the battery box **1002**. In one embodiment, the door is rotatably coupled with the battery box **1002** by a hinge. The door may be configured to open and close by rotating around a hinge axis. The hinge may be on any side of the door. In one embodiment, the hinge is at the top of the door such that the door flips up when opening and down when closing. In other embodiments, the hinge is placed on other sides of the door such that the door opens and closes in different directions. According to an exemplary embodiment, as shown in FIG. **13A**, the battery box **1002** includes only one door such that opening the door can provide access to any module **1106** within the battery box **1002**. In another exemplary embodiment, as shown in FIG. **13B**, the battery box **1002** includes separate doors for each module **1106** (e.g., opening a first door provides access to a first module and opening a second door provides access to a second module) or a subset of a plurality of modules **1106** (e.g., opening a first door provides access to a first module and a second module and opening a second door provides access to a third and fourth module).

(101) According to another exemplary embodiment, as shown in FIG. **13C**, the sealing mechanism **1120** is a locking mechanism integrated with the module **1106**. The locking mechanism may be coupled with an exterior surface of the module **1106** where the exterior surface of the module **1106** creates part of the shell **1102** of the battery box **1002** (e.g., the exterior surface of the module **1106** is flush with the shell **1102** of the battery box **1002**). The locking mechanism may cause, when activated, the exterior surface of the module **1106** to create an airtight seal with the shell **1102** of the battery box **1002**. In some embodiments, the module **1106** has a handle **1302** on the exterior surface. Once positioned appropriately within the battery box **1002**, the handle **1302** may be configured to move from an unlocked position to a locked position. The locked position may be configured to secure the module **1106** in place and create an airtight seal between the module **1106** and the battery box **1002**. The locking mechanism may be configured to release the airtight seal when the handle **1302** is moved from the locked position to the unlocked position.

(102) The sealing mechanisms **1120** as described may be on one, some, or all sides of the battery box **1002**. The sealing mechanisms **1120** may be disposed on the left side **1020**, the right side **1018**, the front side **1012**, or the back side **1014** of the battery box **1002**, or any combination thereof. According to an exemplary embodiment, a module **1106** is accessible from both a first side (e.g., the left side **1020**) and a second side (e.g., the right side **1018**) of the battery box **1002**. In other embodiments, a first module **1106** and a second module **1106** are accessible from the first side of the battery box **1002** and the second side of the battery box **1002**. In another embodiment, the first module **1106** is accessible from the first side of the battery box **1002** and the second module **1106** is accessible from the second side of the battery box **1002**.

(103) According to an exemplary embodiment, the battery box **1002** includes at least one charge port **1122**. The charge port **1122** may be configured to receive an electrical input from an external power source. The electrical power source may be any device capable of transferring energy to another device (e.g., a generator, battery, etc.). The charge port **1122** may be further configured to provide the electrical input to a module **1106** to charge at least one battery **60** of the module **1106** when the module **1106** is disposed within the internal cavity **1104** of the battery box **1002**. As such, instead of having to replace the batteries **60**, the batteries **60** may be recharged. According to an exemplary embodiment, as shown in FIGS. **13A** and **13B**, the battery box **1002** includes a charge port **1122** configured to receive an electrical input from a power source (e.g., a generator) such that a battery **60** can be charged when the module **1106** is disposed within the battery box **1002**. The charge port **1122** may be electrically coupled (either directly or indirectly) with each module **1106** such that energy received by the charge port **1122** may charge any battery **60** within the battery box **1002**. In another embodiment, as shown in FIG. **13C**, the battery box **1002** includes a plurality of charge ports **1122**. A first charge port **1122** may be configured to provide energy to a first module **1106** and a second charge port **1122** may be configured to provide energy to a second module **1106**. In another embodiment, the first charge port **1122** is configured to provide energy to a first subset

of modules **1106** and a second charge port **1122** is configured to provide energy to a second subset of modules **1106**.

(104) According to an exemplary embodiment, as shown in FIGS. **14A-14B**, a battery box **1002** is configured to receive energy from a module **1106** and selectively direct the energy to components of the vehicle **10** (e.g., a tailgate, a lift system). According to an exemplary embodiment, as shown in FIG. **14A**, the battery box **1002** includes at least one power distribution unit **1124**. The power distribution unit **1124** may include a controller **1412**. The controller **1214** may include a processor **1414** and a memory **1416** to facilitate distribution of power to components. For example, the memory **1416** may include instructions that dictate which component to direct power to. The processor **1414** may receive and analyze various inputs and direct power to a component(s) based on the inputs and the instructions. The power distribution unit **1124** may be configured to receive an electrical input from at least one module **1106**. In some embodiments, a first module **1106** is electrically coupled with the power distribution unit **1124** such that energy from the battery **60** disposed in the first module **1106** can be transferred to the power distribution unit **1124**. A second module **1106** may also be electrically coupled with the power distribution unit **1124** such that energy from the battery **60** disposed in the second module **1106** can be transferred to the power distribution unit **1124**. The power distribution unit **1124** may be configured to selectively direct the energy received from the first module **1106** and the energy received from the second module **1106** to at least one component of the vehicle **10** to be operated. According to an exemplary embodiment, the power distribution unit **1124** is configured to direct energy to a tailgate **132** of a refuse vehicle **100**. The power distribution unit **1124** may also be configured to direct energy to a lift assembly **108** of the refuse vehicle **100**. The power distribution unit **1124** may be configured to provide energy to only one component at a time. In other embodiments, the power distribution unit **1124** is configured to provide energy to a plurality of components simultaneously. According to another exemplary embodiment, the battery box **1002** includes a plurality of power distribution units **1124** such that each power distribution unit **1124** can be configured to receive an electrical input from a subset of the plurality of modules **1106**. The subset of the plurality of modules **1106** may include a single module **1106** or several modules **1106**. The power distribution unit **1124** may facilitate the distribution of energy to components of the vehicle **10** even when a module **1106** is out of energy. In some embodiments, when a first module **1106** is out of energy and a second module **1106** and a third module **1106** still have energy, the power distribution unit **1124** directs the energy to the component to be operated.

(105) According to another exemplary embodiment, as shown in FIG. **14B**, the battery box **1002** is configured such that a module **1106** has an associated component (e.g., a component electrically coupled to the module **1106**). Energy from the module **1106** may be directed to the associated component. In some embodiments, a module **1106** provides energy directly to a tailgate **132** of a refuse vehicle **100**. If the module **1106** runs out of energy, the tailgate **132** can no longer operate. In another embodiment, a plurality of modules **1106** are associated with the same component. A first module **1106** and a second module **1106** may be associated with the tailgate **132** of the refuse vehicle **100**. If the first module **1106** runs out of energy, the tailgate **132** can still operate on the energy from the second module **1106**. According to another exemplary embodiment, a module **1106** supplies energy to a plurality of components. The module **1106** may provide energy to the tailgate **132** and a lift assembly **108**. If the module **1106** runs out of energy, the tailgate **132** and the lift assembly **108** can be inoperable, unless associated with a second module **1106**.

(106) According to an exemplary embodiment, as shown in FIG. **15**, method **1200** provides energy to at least one component of a vehicle **10**. At operation **1202**, a chassis **20** of a vehicle **10** receives a module **1106**. At operation **1204**, the chassis **20** transfers energy from the module **1106** to at least one component of the vehicle **10**. At operation **1206**, the chassis **20** receives supplemental energy. In some embodiments, operation **1202** includes a battery box **1002** of the chassis **20** receiving a module **1106**. Receiving the module **1106** may include the module **1106** entering an internal cavity

1104 of the battery box **1002** and positioning the module **1106** within the battery box **1002** such that energy can be transferred from or to the module **1106**. The module **1106** may be disposed within the battery box **1002** and positioned such that a module terminal **1110** of the module **1106** contacts a system terminal **1112** of the battery box **1002**. Contact between the module terminal **1110** and the system terminal **1112** facilitates the transfer of energy from the module **1106** to the battery box **1002** and then to the components of the vehicle **10**. In another embodiment, a first module **1106** with a supplying module terminal **1110** is positioned such that the supplying module terminal **1110** contacts a receiving module terminal **1110** of a second module **1106**. The first module **1106** and the second module **1106** may be connected in series such that energy from the first module **1106** can transfer through the second module **1106** before transferring to the battery box **1002** or to a third module **1106**. According to an exemplary embodiment, the second module **1106** includes a supplying module terminal **1110** to contact at least one of a receiving module terminal **1110** of a third module **1106** and the system terminal **1112** of the battery box **1002**. A mounting provision **1114** may facilitate the positioning of the module **1106** within the battery box **1002**. The mounting provision **1114** may be a track or a wall along which the module **1106** can slide when entering, or leaving, the battery box **1002**.

(107) In some embodiments, receiving the module **1106** at operation **1202** includes locking the module **1106** in an appropriate position. The battery box **1002** may include a mounting provision **1114** configured to keep the module **1106** stationary relative to the battery box **1002**. The mounting provision **1114** may be any structural element that limits the movement of the module **1106** when disposed in the internal cavity **1104** of the battery box **1002**. According to an exemplary embodiment, the mounting provision **1114** is a ridge beyond which the module **1106** cannot slide. In other embodiments, the mounting provision **1114** is a receiver configured to receive a pin coupled with the module **1106** when aligned properly or a movable member configured to move between a loading position and a locking position, among others.

(108) According to an exemplary embodiment, at operation **1204** the chassis **20** transfers energy from the module **1106** to at least one component of the vehicle **10**. The transfer of energy may be direct from the module **1106** to the component, or it can be indirect. In some embodiments, a module **1106** is electrically coupled with a component via the battery box **1002** of the chassis **20** such that energy from the module **1106** can transfer directly to the component when operated (e.g., energy from the module **1106** is directly transferred to an engine of the vehicle **10**). In another embodiment, the module **1106** is coupled with a second module **1106** such that energy is transferred through the second module **1106** before reaching the component, or the module **1106** is coupled with a power distribution unit **1124** such that energy is transferred to the power distribution unit **1124** before reaching the component. The battery box **1002** of the chassis **20** may include system terminals **1112** to facilitate the transfer of energy from the module **1106** to the component of the vehicle **10**. In some embodiments, a module **1106** includes a module terminal **1110** that contacts the system terminal **1112**. The module terminal **1110** and the system terminal **1112** may comprise materials capable of conducting electricity and transferring energy. The system terminal **1112** may be electrically coupled to a component of the vehicle **10** or a power distribution unit **1124**, or any combination thereof, such that the energy can reach the component to be operated.

(109) According to an exemplary embodiment, at operation **1206** the chassis **20** receives supplemental energy. The supplemental energy may be energy from the same module **1106** or a different module **1106**. In some embodiments, a battery **60** within the same module **1106** is recharged to receive supplemental energy from the same module **1106**, or the battery **60** is replaced. A battery box **1002** of the chassis **20** may include a charge port **1122** such that the battery **60** disposed in the module **1106** can remain in the module **1106**, and the module **1106** can remain in the battery box **1002**, when the battery **60** is being recharged. The battery box **1002** may include a charge port **1122** configured to recharge a single module **1106** or a plurality of modules **1106**. In some embodiments, a first charge port **1122** is configured to provide energy to a first module **1106**,

and a second charge port **1122** is configured to provide energy to a second module **1106**. In another embodiment, the first charge port **1122** is configured to provide energy to the first and the second module **1106**, and the second charge port **1122** is configured to provide energy to a third and a fourth module **1106**. In another embodiment, one charge port **1122** is configured to supply energy to all modules **1106** disposed within the battery box **1002**. In yet another embodiment, instead of recharging the battery **60** within the module **1106**, a first battery **60** originally disposed in the module **1106** is replaced with a second battery **60**. The module **1106** may be removed from the battery box **1002** such that the first battery **60** within the module **1106** may be accessed. The first battery **60** may be replaced by the second battery **60** and then the same module **1106** may be reinserted into the battery box **1002**.

(110) According to another exemplary embodiment, to receive the supplemental energy from a different module **1106**, a first module **1106** disposed within the battery box **1002** is replaced with a second module **1106**. The second module **1106** may include at least one battery **60** with energy ready to be transferred to a component of the vehicle **10**. The battery box **1002** of the chassis **20** may be configured to incorporate any combination of recharging batteries **60**, replacing batteries **60**, and replacing modules **1106**.

Structural Body

(111) According to the exemplary embodiments shown in FIGS. **19-25**, a vehicle, shown as vehicle **1500**, is configured as an electric vehicle as described above. The vehicle **1500** may be similar to the vehicle **10** described herein, with similar features identified using like reference numerals, except as described herein or as apparent in the figures. The vehicle **1500** includes one or more electrical energy storage devices, shown as a battery housing **1502** and one or more batteries **60**, positioned within middle section **24**. The vehicle **1500** further includes one or more electromagnetic drive motors (not shown) coupled to the batteries **60** to propel the vehicle **1500**. The vehicle **1500** may also be configured with the any of the application kits **80** described herein, provided the application kit **80** serves as the main structural component of vehicle **1500** by connecting front section **22** to rear section **26**.

(112) According to an exemplary embodiment shown in FIG. **19**, the vehicle **1500** includes a chassis, shown as chassis **20**, and a cab, shown as cab **40**, connected to a structural body, shown as structural body **1510**. In some embodiments the vehicle **1500** is the same as vehicle **10** above. In some embodiments, the cab **40** is coupled directly with the structural body **1510**. In some embodiments, the cab **40** and the structural body **1510** are an integral, unitary piece. Still in other embodiments, the cab **40** is connected to the structural body **1510** indirectly, for example through the front section **22**. The structural body **1510** may include frame rails, such as C-shaped frame rails, cross members, or other structural components and/or members to provide structural support to the vehicle **1500**. For example, the structural body **1510** can act as a conventional frame member and can include frame portions, frame members, or frame rails, etc., necessary to provide structural support to the vehicle **1500** and its various components. In some embodiments, the structural body **1510** can provide structural support without frame rails. The structural body **1510** may also act as the application kit **80** and provide the vehicle **1500** with various configurations. In such embodiments, the structural body **1510** provides structural support to the vehicle **1500** as the vehicle's main stressed member and includes one or more implements, vehicle bodies, and/or other components for the variety of the application kits **80** described above. For example, when the vehicle **1500** is configured as a refuse truck as shown in FIGS. **3** and **19**, structural body **1510** can include a refuse compartment and loading portion for facilitating the transport of refuse, while also providing structural support to the vehicle as the main stressed member between front and rear components. For another example, referring back to FIG. **5**, the vehicle **10** is configured as a fire fighting vehicle, and the application kit **80** includes deployable stabilizers, storage compartments, the ladder assembly **308**, etc. According to the present disclosure, in some embodiments the structural body **1510** includes the application kit **80** and also acts as main structural member for the

fire fighting vehicle **300**, connecting chassis components such as the front section **22** and the rear section **26** to provide longitudinal, lateral, and twisting strength to the fire fighting vehicle **300** as the main structural member. The application kit **80** as the main stressed member removes the stressed members traditionally found in the middle section **24** of the chassis **20** and allows the vehicle **300** to carry more components, such as the batteries **60** for configuring the fire fighting vehicle **300** as an electrified vehicle.

(113) According to an exemplary embodiment shown in FIG. **20**, the structural body **1510** is the main stressed member of the vehicle **1500**. In some embodiments, the structural body **1510** is part of the chassis **20**. For example, the structural body **1510** can be a part of middle section **24** and serve to connect front section **22** to rear section **26**. In some embodiments, the structural body **1510** connects one or more components of the vehicle **1500**. In some embodiments, the structural body **1510** is a stressed member that replaces a conventional frame. The structural body **1510** can eliminate redundant components and can reduce the weight of the vehicle **1500**. The structural body **1510** may also minimize the space required on-vehicle for structural members.

(114) Still referring to the exemplary embodiment shown in FIG. **20**, the structural body **1510** connects components of the chassis **20** including the front section **22**, one or more other components of the middle section **24** (shown to include the battery housing **1502** and the batteries **60**), and the rear section **26**. In some embodiments, the front section **22** and the rear section **26** are subframes as described above. Other components of the vehicle may connect to one or more of the structural body **1510**, the front section **22**, and the rear section **26**, such as suspension components, drive train components, steering components, etc. The structural body **1510** connects the front section **22** and the rear section **26** and allows forces to be passed between the front section **22** and the rear section **26**. For example, in a front-wheel drive configuration of vehicle **1500**, driving force from the front tire assemblies **54** can be transferred to the rear section **26** via the structural body **1510**. The structural body **1510** connects to the front section **22** and the rear section **26** by bolts or other fastening means (e.g., welds) to thereafter act as the main stressed member. In some embodiments, the structural body **1510** is integral with the front section **22** and the rear section **26** as a single stressed member. The structural body **1510** as described may replace a conventional frame/subframe arrangement as the structural member(s) between front and rear axles. The front section **22**, the rear section **26**, and the structural body **1510** act as the main structural member and transfer stresses within the vehicle **1500**. The structural body **1510** may include the same mounting points in the same position as the frame in middle section **24** when middle section **24** serves as a frame member connecting the front section **22** and the rear section **26**. The structural body **1510** thereafter replaces the middle section **24** as the connecting member between the front section **22** and the rear section **26**.

(115) The front section **22** and the rear section **26** may provide one or more hard-point mounting provisions for suspension components (e.g., steering arms, steering actuators, etc.), steering components (e.g., gas springs, dampeners, air springs, etc.), and/or power components (e.g., differentials, drive shafts, etc.) of the vehicle **1500**. The front section **22** and rear section **26** may also include various other components of vehicle **10** described above. For example as shown in FIG. **1**, the rear section **26** can include the drive motor **62**, and as shown in FIG. **19** the front section **22** can include the front axle **50** and the rear section **26** can include one or more the rear axles **52**. The front section **22** and the rear section **26** also include one or more tire assemblies **54** for propelling the vehicle **1500**.

(116) According to an exemplary embodiment shown in FIG. **21**, the middle section **24** includes the space between the front section **22** and the rear section **26**, where a vehicle's frame conventionally is found. According to the exemplary embodiments described herein, the structural body **1510** acts as the main structural member connecting the front section **22** and the rear section **26**, as such the vehicle **1500** does not require the middle section **24** to be a structural component. Accordingly, the middle section **24** may house other vehicle components.

(117) According to the exemplary embodiments shown in FIGS. 19-25, the middle section 24 houses an energy storage system for powering electric motors of vehicle 1500, shown as batteries 60 and battery housing 1502. The battery housing 1502 protects the batteries 60 from hazards including water, salt, dust, mud and debris. In some embodiments, the battery housing 1502 thermally isolates the batteries 60 from the outside environment. The batteries 60 may be contained in a tray or enclosure within the battery housing 1502. In some embodiments, the battery housing 1502 has integrated features for securing the batteries inside. For example, the tray may include indexed spaces for battery components. The batteries 60 may be removable from the battery housing 1502 for repair and/or replacement. In some embodiments, batteries 60 can be removed from battery housing 1502 while battery housing 1502 is coupled to structural body 1510. The batteries may be replaced and reinstalled without disconnecting the battery housing 1502 for more efficient servicing.

(118) Referring back to the exemplary embodiment shown in FIG. 20, the battery housing 1502 is a rectangular assembly. In other embodiments, the battery housing 1502 may be another shape configured to fit within the available space between the front section 22 and the rear section 26 while maintaining the ability to access and remove the battery housing 1502 from the side of the vehicle 1500 between the front section 22 and the rear section 26. Still in other embodiments, the battery housing 1502 may extend beyond the front axles or the rear axles. The battery housing 1502 may include one or more selectively removable panels. For example a front panel, shown as front panel 1506, is removable and provides access to batteries 60 and other systems within battery housing 1502. In other embodiments, the battery housing 1502 includes one or more other access points (not shown) providing access to its interior and the batteries 60. The access points can be doors, latches, removable panels, etc., that can be sealed to prevent hazards from entering the interior of the battery housing 1502. The battery housing 1502 includes batteries 60 as well as associated systems and components such as cabling, thermal management systems, charging systems, communication systems, protection systems, etc. In some embodiments, the thermal management system includes an insulating layer that substantially surrounds the batteries 60 in the battery housing 1502. In some embodiments, the thermal management system can include a heating element and/or a cooling element. In some embodiments, the thermal management system includes one or more fluid channels in the battery housing 1502 for coolant to flow. In some embodiments, the thermal management system includes one or more heat exchangers, pumps, and radiators to heat and/or cool the coolant. In some embodiments, the thermal management system includes a controller. The controller can include a processing circuit including a processor and a memory. The processing circuit can be communicably connected to a communication interface to communicate with the one or more systems of the vehicle 1500 and/or systems external to the vehicle 1500. The processor can be implemented as a general purpose processor, an application specific integrated circuit ("ASIC"), one or more field programmable gate arrays ("FPGAs"), a group of processing components, or other suitable electronic processing components.

(119) The memory (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. The memory can be or include volatile memory or non-volatile memory. The memory can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, the memory is communicably connected to the processor via the processing circuit and includes computer code for executing (e.g., by the processing circuit and/or the processor) one or more processes described herein.

(120) In some embodiments, the controller for the thermal management system is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments, the

functionality of the controller can be distributed across multiple servers or computers (e.g., that can exist in distributed locations).

(121) In some embodiments, a charging system includes a charging circuit, which can include one or more transformers, rectifiers, capacitors, or contactors. In some embodiments, the charging circuit is connected to a charging outlet of the vehicle to receive power from an external power source. In some embodiments, the charging circuit is electrically connected to the drive motor **62** to receive power, for example during recharging events where the vehicle uses the drive motor **62** to generate power. In some embodiments, the aforementioned systems are assembled with the batteries **60** and installed in the battery housing **1502** with the batteries **60**. The battery housing **1502** contains the batteries **60** and related systems and may act as a single modular component readily installable and removable from the vehicle **1500**.

(122) In some embodiments, the batteries **60** can be a series of sections, battery strings, battery pods, battery cells, battery modules, or battery assemblies. The batteries **60** provide power for electric drive motors of the vehicle **1500** (e.g., drive motors **62** of vehicle **10**) via one or more electrical connections. In some embodiments, the batteries **60** also provide power for the other components of the vehicle **1500** such as one or more implements included in the application kit **80**. For example, the vehicle **1500** may include a vehicle power system (not shown) for distributing power throughout the vehicle **1500**. The vehicle power system may electrically connect the electric motors of the vehicle **1500** to the battery housing **1502** and the batteries **60** via an electrical connection, shown as electrical connection **1504**. The vehicle power system may include wires, processors, memory, and user input devices. While only one electrical connection is shown, multiple electrical connections be provided. The vehicle power system may be integrated into the one or more components that facilitate user operation of the vehicle **1500**, and may integrate with user interface components in the cab **40** or elsewhere on the vehicle **1500**. In some embodiments, the electrical connection **1504** can be a plug, an outlet, an electrical contact, a bus bar connection, or any other kind of connection suitable for electrically connecting the batteries **60** to the vehicle power system of the vehicle **1500** in a selectively detachable manner. In some embodiments, the electrical connection **1504** automatically connects the battery housing **1502** and the vehicle power system as the battery housing **1502** is installed. In other embodiments, the electrical connection **1504** is connected after the battery housing **1502** is installed in the vehicle **1500**.

(123) Referring back to the exemplary embodiment shown in FIG. **19**, the battery housing **1502** is protected by impact barrier(s), shown as impact barriers **1520** on each side of the vehicle **1500**. The impact barriers **1520** protect the middle section **24** of the vehicle **1500**, including the battery housing **1502** and the batteries **60**, from side impacts, road debris, and other dangers. In some embodiments, the impact barriers **1520** are on opposing sides of the battery housing **1502**. For example, an impact barrier **1520** can face a first side of the battery housing **1502** and an impact barrier **1520** can face a second side of the battery housing **1502** opposite the first side. The first and second sides can be lateral sides of the battery housing **1502**, with one lateral side on a left side of the vehicle **1500** and a lateral side on the right side of the vehicle **1500**.

(124) The impact barriers **1520** may be selectively removable allowing access to battery housing **1502** and batteries **60** from the side of vehicle **1500**. While shown in FIG. **19** as a plate, in some embodiments, the impact barriers **1520** can rails, bumpers, and/or other arrangements configured to protect the middle section **24**. The impact barriers **1520** may be attached with latches, bolts, hinges, sliders, or other mechanisms that facilitate sliding, rotating, or removing the impact barriers **1520** to allow access to the battery housing **1502**.

(125) According to the exemplary embodiment shown in FIG. **21**, the impact barriers **1520** can be removed to provide access to the battery housing **1502**. In some embodiments, the battery housing **1502** may selectively couple and decouple from the structural body **1510**. In some embodiments, the battery housing **1502** is permanently coupled to the structural body **1510** and instead the batteries **60** may selectively couple and decouple from the battery housing **1502**. In such

embodiments, the batteries **60** include the same connections and features of a selectively removable battery housing **1502** as described herein. In some embodiments, the battery housing **1502** can include integrated fork lift guides, shown as lift guides **1550** for facilitating the installation and removal of the battery housing **1502** from the vehicle **1500**. While shown as lift guides **1550**, other embodiments of vehicle **1500** may include lift points, slots, tracks, etc.

(126) As shown in FIG. **21**, the structural body **1510** includes one or more installation guides, shown as guide pins **1562**, for properly positioning the battery housing **1502** in relation to the structural body **1510** during installation. In some embodiments, the guide pins **1562** correspond to exclusions, indentations, and/or corresponding receiving portions in battery housing **1502** that receive the guide pins **1562**. While shown as guide pins **1562**, the guide pins **1562** may be wedges, indents, indexes, etc., configured to extend from the structural body **1510** and interface with the battery housing **1502**. In some embodiments, the guide pins **1562** position the battery housing **1502** appropriately so that the one or more mounting points of the battery housing **1502** align with corresponding mounting points on structural body **1510**. The guide pins **1562** may position the battery housing **1502** both in the proper longitudinal and lateral position, as well as in a parallel plane to the mounting points of structural body **1510**. As discussed herein the guide pins **1562** may also provide engagement limits or stops for translational and/or rotational movement of the battery housing **1502** relative to the structural body **1510**.

(127) Still referring to the exemplary embodiment shown in FIG. **21**, one or more mounting points, shown as mounting points **1530** and center mounting points **1540**, can couple the battery housing **1502** to the structural body **1510**. In some embodiments, the mounting points **1530** and center mounting points **1540**, either individually or as a group, provide torsional and bending isolation of the battery housing **1502** from the structural body **1510**. The isolation protects the battery housing **1502** and the batteries **60** as the structural body **1510** is stressed, during which it may experience some deformation. In some embodiments, the torsional and bending isolation is provided by springs, dampers, etc. For example, at least a portion of the mounting points **1530** and center mounting points **1540** can be made of rubber to dampen forces exchanged between the structural body **1510** and the battery housing **1502**. As shown in FIGS. **21** and **22**, the battery housing **1502** is mounted to the structural body **1510** using a four-point mounting system including mounting points **1530** and center mounting point(s) **1540**.

(128) According to an exemplary embodiment shown in FIG. **22**, as viewed from below the vehicle **1500**, the center mounting points **1540** are pivot mounts that allow the battery housing **1502** to rotate around the vehicle **1500**'s lateral axis, shown as lateral axis **1570**. The mounting points **1530** are also pivot mounts that allow the battery housing **1502** to rotate around the vehicle **1500**'s longitudinal axis, shown as longitudinal axis **1575**.

(129) According to an exemplary embodiment shown in FIG. **23**, a three-point mounting system, shown as fixed points **1555** and pivoting mount **1565**, connects the battery housing **1502** to the structural body **1510**. The two fixed points **1555** are located near the rear of battery housing **1502** and the single pivoting mount **1565** is located near the front of battery housing **1502**. The three-point mounting system may be reversed such that the fixed points **1555** are nearer the front of the vehicle **1500**. The pivoting mount **1565** allows the battery housing **1502** to rotate around an axis parallel to the longitudinal axis of the vehicle **1500**, shown as longitudinal axis **1585**, as well as an axis parallel to the lateral axis of the vehicle, shown as lateral axis **1580**. While the three-point and four-point mounting systems are shown, in FIGS. **12** and **13**, respectively, it should be understood that other mounting systems and arrangements may be used to couple the battery housing **1502** and the batteries **60** to the structural body **1510** while isolating the battery housing **1502** from any torsional or bending forces. The mounting points **1530**, center mounting points **1540**, fixed points **1555**, and pivoting mount **1565** may additionally isolate battery housing **1502** from any vibrational forces in the structural body **1510**.

(130) As shown in FIGS. **22** and **23**, the battery housing **1502** may extend the width of the vehicle

1500 along a lateral axis, such as lateral axis **1570**, and along the longitudinal axis of the vehicle **1500**, such as longitudinal axis **1575**. As defined herein the width of the vehicle **1500** is the distance between the outermost surfaces of tire assemblies **54** along the lateral axis of the vehicle **1500**. In some embodiments, the battery housing **1502** extends between the front axle **50** and the rear axle(s) **52**. In some embodiments battery housing **1502** extends beyond front axle **50** and/or rear axle(s) **52**. The length, width, and height of battery housing **1502** may vary depending on the desired or required size of the batteries **60** and of vehicle **1500**.

(131) According to an exemplary embodiment in FIG. **24**, the middle section **24** includes the battery housing **1502** and the electrical connection **1504**. The battery housing **1502** includes the installation guides, shown as installation guides **1560**, that receive the guide pins **1562** in order to facilitate installation and proper positioning of mounting points with the structural body **1510**, shown as the mounting points **1530** and the center mounting points **1540**. As shown, the guide pins **1562** are near each corner of the battery housing **1502**, though it should be understood the guide pins **1562** can be positioned anywhere between the structural body **1510** and the battery housing **1502**. In some embodiments, the guide pins **1562** support the battery housing **1502** in both the x and y planes parallel to the plane of the vehicle **1500**. Cross-section A is indicated through the middle of the battery housing **1502** and is shown in FIG. **25**.

(132) According to the exemplary embodiment shown in FIG. **25**, the installation guides **1560** are tapered exclusions in the battery housing **1502** to receive corresponding the guide pins **1562**. While shown on the top side of the battery housing **1502**, the installation guides **1560** and/or other types of installation guides may be located on other faces of the battery housing **1502**, such as the sides. While four installation guides **1560** are shown, any number of installation guides, indexes, alignment pins, etc., can be used to align the battery housing **1502** with the mounting points on the structural body **1510**.

(133) Although this description may discuss a specific order of method steps, the order of the steps may differ from what is outlined. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(134) As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(135) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(136) The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a

single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(137) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(138) The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

(139) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(140) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above.

Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(141) It is important to note that the construction and arrangement of the vehicle **10** as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. For example, the cab **40** of the exemplary embodiment shown in at least FIG. **4** may be incorporated into the refuse vehicle **100** of the exemplary embodiment shown in at least FIG. **3**. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

Claims

1. An electrified vehicle, comprising: a front subframe; a rear subframe; a body that connects the front subframe to the rear subframe so that forces acting on the front subframe and the rear subframe are transmitted through the body; a housing coupled to an underside of the body between the front subframe and the rear subframe; and an energy storage system enclosed within the housing and comprising: a battery; and a battery interface configured to electrically couple the battery to at least one component of the electrified vehicle, wherein the body comprises at least one of a refuse compartment, a lift assembly, a mixing drum assembly, a ladder assembly, a pump system, or a turntable.
2. The electrified vehicle of claim 1, further comprising: a front tractive element attached to the front subframe; a rear tractive element attached to the rear subframe; and a drive motor electrically coupled to the energy storage system configured to drive at least one of the front tractive element or the rear tractive element.
3. The electrified vehicle of claim 1, wherein the body is directly coupled to the front subframe and the rear subframe such that stresses are transferred between the front and rear subframes.
4. The electrified vehicle of claim 1, wherein the body comprises a refuse compartment and a lift assembly configured to selectively extend from the body.
5. The electrified vehicle of claim 1, further comprising: a plurality of barriers coupled to the body, wherein at least one of the plurality of barriers faces a first side of the housing and at least one of the plurality of barriers faces a second side of the housing opposite the first side, wherein at least one of the plurality of barriers is movable to provide access to at least one of the first side or the second side of the energy storage system.
6. The electrified vehicle of claim 1, wherein the housing is coupled to the body via a three-point connection or a four-point connection.
7. The electrified vehicle of claim 1, wherein the housing is coupled to the body via at least one pivot point at least partially composed of rubber.
8. The electrified vehicle of claim 1, wherein the mounting system comprises: a first plurality of pivot mounts rotatably coupling the housing to the body such that the housing is rotatable around a first axis; and a second plurality of pivot mounts rotatable coupling the housing to the body such that the housing is rotatable around a second axis orthogonal to the first axis.
9. The electrified vehicle of claim 1, wherein the mounting system comprises: a pivot mount rotatably coupling the housing to the body such that the housing is rotatable around a longitudinal axis; and a plurality of fixed mounts coupling the housing to the body, wherein the plurality of fixed mounts are spaced longitudinally a distance apart from the pivot mount with at least one of

the plurality of fixed mounts on a left side of the electrified vehicle and at least one of the plurality of fixed mounts on a right side of the electrified vehicle.

10. An electrified vehicle comprising: a front subframe; a rear subframe; a body that connects the front subframe to the rear subframe so that forces acting on the front subframe and the rear subframe are transmitted through the body; a housing coupled to an underside of the body between the front subframe and the rear subframe; an energy storage system enclosed within the housing and comprising: a battery; and a battery interface configured to electrically couple the battery to at least one component of the electrified vehicle, a first barrier removably coupled to the body on a first lateral side of the housing; and a second barrier removably coupled to the body on a second lateral side of the housing opposite the first lateral side, the first and second barriers positioned longitudinally between the front subframe and the rear subframe.

11. The electrified vehicle of claim 10, wherein at least one of the first barrier and the second barrier is removeable to provide access to at least one of the first lateral side or the second lateral side of the housing.

12. An electrified vehicle, comprising: a chassis comprising: a front subframe; a rear subframe; and a body that directly couples the front subframe to the rear subframe; a tractive element supported by the chassis; a drive motor supported by the chassis and coupled to the tractive element to rotate the tractive element; a housing coupled to the body between the front subframe and the rear subframe, the housing comprising a battery, wherein the battery is electrically coupled to the drive motor; a first wall coupled to the body on a first lateral side of the housing; and a second wall coupled to the body on a second lateral side of the housing opposite the first lateral side, the first and second walls positioned longitudinally between the front subframe and the rear subframe, wherein at least one of the first wall and the second wall is movable to provide access to at least one of the first side or the second side of the housing.

13. The electrified vehicle of claim 12, wherein the housing is removably coupled to an underside of the body between the front subframe and the rear subframe.

14. The electrified vehicle of claim 12, wherein the body comprises at least one of a refuse compartment, a lift assembly, a mixing drum assembly, a ladder assembly, a pump system, or a turntable.

15. The electrified vehicle of claim 12, wherein the body comprise a refuse compartment and a lift assembly configured to selectively extend from the body.

16. The electrified vehicle of claim 12, wherein the housing is removable from the body when the body is coupled to the front subframe and the rear subframe.

17. The electrified vehicle of claim 12, wherein body directly couples the front subframe to the rear subframe so that forces acting on the front subframe and the rear subframe are transmitted through the body.

18. The electrified vehicle of claim 12, wherein the housing is removably coupled to the body via a plurality of attachment points, wherein each of the plurality of attachment points is accessible from an exterior of the electrified vehicle.

19. The electrified vehicle of claim 12, wherein the housing is coupled to the body via a first pivot mount and a first fixed mount.

20. An electrified vehicle, comprising: a chassis comprising: a front subframe; a rear subframe; and a body that directly couples the front subframe to the rear subframe; a tractive element supported by the chassis; a drive motor supported by the chassis and coupled to the tractive element to rotate the tractive element; a housing coupled to the body between the front subframe and the rear subframe, the housing comprising a battery, wherein the battery is electrically coupled to the drive motor, wherein the housing is coupled to the body via a first pivot mount and at least one of a second pivot mount or a first fixed mount.
